
Stable Rank Based Analysis of Low-Rank Adaptation: Tighter Expressivity and Generalization Bounds

Nikita Okhotnikov
okhotnikov.nv@phystech.edu
Moscow Institute of Physics and Technology

Low-rank adaptation (LoRA) has become the de facto standard for parameter-efficient fine-tuning of large language models. Existing theoretical bounds on expressivity and generalization of LoRA are formulated in terms of algebraic rank, which fails to capture the empirical observation that trained matrices often exhibit stable rank significantly lower than their nominal rank. In this paper, we propose stable rank as a more appropriate theoretical measure for analyzing LoRA’s properties. We derive fundamental limits on single matrix approximation under stable rank constraint, extend these to expressivity bounds for fully connected networks and transformers. Additionally, we prove a tighter generalization bound proportional to $\sqrt{\text{sr}(\Delta W)}$ rather than previous best $\sqrt{\text{rank}(\Delta W)}$ from the literature, using a PAC-Bayesian framework. Our empirical validation additionally confirms that LoRA exhibits implicit bias toward low stable rank solutions both in small scale toy setup and on actual downstream tasks with pretrained LLM. We also validate our theoretical findings and demonstrate via controlled experiments on synthetic data that $\mathcal{O}(\sqrt{\text{sr}(\Delta W)})$ is more practical estimate for generalization gap.

Содержание

1	Introduction	2
2	Theoretical Background	3
2.1	Parameter-Efficient Fine-tuning	3
2.2	LoRA Theoretical Background	4
2.3	Stable Rank in Machine Learning	4
2.4	PAC-Bayesian Framework	5
3	Mathematical Framework	6
3.1	Preliminaries	6
3.2	Single Matrix Approximation	6
3.3	Fully Connected Network Approximation	9
3.3.1	Notation	9
3.3.2	Expressive power theorem	9
3.4	Transformer Network Approximation	11
3.4.1	Notation	11

3.4.2	Expressive power of transformer network	12
3.5	Generalization Analysis	16
3.5.1	PAC-Bayesian Framework Integration	16
3.5.2	Notation	17
3.5.3	Upper-bound for LoRA Generalization Gap	17
4	Empirical Validation	22
4.1	Toy experiment details	22
4.2	LLM Experiments	22
4.3	Analysis	22
5	Discussion	22
6	Conclusion	22

1 Introduction

Recent advances in large language models (LLMs) and multimodal models (MLLMs) induce a growing number of various downstream tasks attempted to be solved with a single pretrained model. Despite the fact that the performance of such models has vastly improved over recent years, there are still tasks that cannot be handled in a zero-shot setting and others that heavily benefit from additional fine-tuning. However, the size of LLMs with cutting-edge performance keeps growing, and even the best open-sourced models nowadays contain hundreds of billions of parameters [4, 34, 45, 7]. Thus, full fine-tuning of such models is computationally infeasible in many cases, making parameter-efficient fine-tuning (PEFT) methods strictly required. Among these, low-rank adaptation (LoRA) [11] has emerged as the most widely adopted technique, enabling the efficient adaptation of billion-parameter models by freezing pretrained weights and injecting small low-rank rank trainable matrices. Since the release of the original paper, the approach has been actively studied [3, 1, 48, 15, 12, 22], yet it still lacks a comprehensive theoretical foundation explaining its expressivity, generalization, expected optimization behavior and some empirically observable effects.

Most previous theoretical results on LoRA are based on the nominal algebraic rank of the adapters as the primary bounding parameter. However, such rank-based approaches diverge from empirical observations: trained LoRA adapters often produce very specific matrices that exhibit a stable rank close to 1 [18] on various downstream tasks. The stable rank of a matrix $A \in \mathbb{R}^{m \times n}$ is defined as:

$$\text{sr}(A) = \frac{\|A\|_F^2}{\|A\|_{op}^2} = \frac{\sum_{i=1}^{\min(m,n)} \sigma_i^2(A)}{\sigma_1^2(A)}, \quad (1)$$

where $\sigma_i(A)$ are the singular values in descending order. This quantity, introduced by Rudelson and Vershynin [31], provides a continuous measure of the “effective dimensionality” of a matrix. Unlike algebraic rank, stable rank is robust to small perturbations and captures the distribution of energy across singular values. The discrepancy between theoretical bounds based on algebraic rank and empirical observations comes from a fundamental mismatch in how we estimate LoRA’s capacity. While algebraic rank provides an upper bound on the number of independent directions a matrix can span, it fails to capture how the gradient-based optimization procedure actually works with these parameters’ theoretical capacity.

In this work, we claim that stable rank is a more mathematically appropriate measure for the theoretical analysis of low-rank adapters, and we formally study the expressivity and generalization properties of the trained adapter under this lens. We additionally confirm the implicit bias of low stable rank in both controlled synthetic experiments and real-world tasks. Unlike previous works, we operate strictly under a stable rank constraint and focus on the dependency of LoRA’s properties on the resulting stable rank. We prove a fundamental lemma on the irreducible approximation error of a single matrix by an estimate of limited stable rank, then providing an actual tight lower bound and the way to construct the matrix that reaches

that bound similar to the Eckart-Young-Mirsky theorem [6]. Using this lemma, we derive a novel, more realistic expressivity upper bound that scales with stable rank rather than algebraic rank for both multi-layer feed-forward networks (FNNs) and slightly simplified transformer [39] architecture. Our results incorporate this implicit optimization bias into previous findings on expressivity limits.

Additionally, we demonstrate that the degree of overfitting – or the size of the generalization gap – is better characterized as a function of stable rank rather than nominal rank. Utilizing a PAC-Bayesian framework [24, 23], we provide a rigorous proof for a generalization gap upper bound that scales as $\mathcal{O}(\sqrt{\text{sr}(\Delta W)/N})$. This significantly tightens the previously known best bound of $\mathcal{O}(\sqrt{\text{rank}(\Delta W)/N})$ [15].

The root cause of the observed low stable rank implicit bias is beyond the scope of this paper, as is an exhaustive analysis of the optimization trajectory and procedure itself. This remains an active subject for future research.

2 Theoretical Background

2.1 Parameter-Efficient Fine-tuning

Parameter-Efficient Fine-tuning (PEFT) has emerged as a crucial paradigm in modern deep learning, designed to adapt massive pre-trained models to downstream tasks while updating only a minimal subset of parameters. The landscape of PEFT methods has diversified significantly over time, with each method offering different trade-offs between parameter efficiency, training speed, inference latency, and task performance.

Before the dominance of LoRA, the primary methods broadly fell into architectural additions and input modifications. Housby et al. [10] introduced bottleneck adapters that insert small MLP layers between existing transformer blocks, while Rebuffi et al. [30] explored learning multiple adapters for multi-task scenarios. These methods achieve good parameter efficiency but fundamentally alter the forward pass, adding unavoidable inference latency due to the sequential nature of the additional layers. Alternatively, Li et al. [17] proposed prefix-tuning that optimizes continuous prefixes of input sequences, while Lester et al. [16] introduced prompt tuning with learned soft prompts. Qin et al. [29] provided a unified framework analyzing the relationship between these prompt-based approaches. While these methods modify inputs rather than weights, avoiding most of the inference latency, they consume valuable context window space and present different theoretical limitations.

LoRA [11] established a new standard by hypothesizing that changes in model weights during fine-tuning have a low "intrinsic dimension." By freezing the pre-trained weights $W_0 \in \mathbb{R}^{d \times k}$ and injecting trainable rank decomposition matrices $A \in \mathbb{R}^{r \times k}$ and $B \in \mathbb{R}^{d \times r}$ such that the update is $\Delta W = BA$, LoRA avoids both additional inference latency (as weights can be merged into pretrained models after training) and context window reductions. Additionally, as some other methods, it allows learning and inferencing multiple adapters for different tasks without re-hosting and resaving the entire pretrained model.

Because of its structural simplicity and lack of inference overhead, LoRA has become the de facto standard PEFT method. Comprehensive comparative studies by Karimi et al. [14] and He et al. [9] empirically verified its strong performance across varying tasks and model scales. Extensive surveys by Ding et al. [5], Han et al. [8], and Xu et al. [44] further document its ubiquity across the PEFT varieties.

Following its success, numerous structural and optimization-based variants have emerged to address specific limitations of the vanilla formulation. Zhang et al. [49] developed AdaLoRA to dynamically allocate rank budgets across different layers based on importance scores, while Valipour et al. [38] introduced DyLoRA for dynamic, search-free rank selection during training. To improve the learning dynamics, Liu et al. [21] proposed LoRA+, which applies asymmetric learning rates to the A and B matrices to overcome initialization-induced optimization inefficiencies. Kalajdzievski [13] introduced Rank-Stabilized LoRA (RS-LoRA), which modifies the scaling factor from $1/r$ to $1/\sqrt{r}$ to mathematically stabilize the learning trajectory and prevent gradient collapse at high ranks. Liu et al. [20] proposed Weight-Decomposed Low-Rank Adaptation (DoRA), which decouples the pretrained weights into magnitude and directional components, applying the standard LoRA update strictly to the directional matrix to closer mimic full fine-tuning behavior. Meng et al. [25] introduced PiSSA, which initializes the A and B matrices using the Principal Singular Values and Vectors of the base model weights instead of random noise, allowing the adapter to immediately optimize the most significant spectral components.

Despite these seeming diversity – whether a method alter the initialization (PiSSA), decouple the weight vectors (DoRA), stabilize the scaling (RS-LoRA), or adjust the learning rates (LoRA+) – all of them

ultimately preserve the core low-rank multiplicative structure $\Delta W = BA$ applied additively to the base weights. Because our mathematical framework operates directly on these matrices and the stable rank of the resulting perturbation matrix ΔW , rather than relying on assumptions about the specific optimizer state or initialization scheme, the theoretical principles and bounds presented in this paper remain fully applicable to these variants.

2.2 LoRA Theoretical Background

The theoretical understanding of LoRA has developed along several complementary directions, though most share a common limitation in their reliance on the algebraic rank parameter r .

Regarding expressive power, Zeng and Lee [48] established a foundational existence proof showing that rank- r LoRA adapters can represent any target perturbation of a given width and depth when certain rank threshold is met. However, this existence result operates purely in the algebraic capacity space, it provides no insight into the actual penalty incurred if gradient-based optimization converges to matrices with skewed singular value distributions or any insights onto the form of this distributions in the «optimal» solution.

From a generalization perspective, Kratsios et al. [15] derived sample complexity and generalization bounds specifically for asymmetric low-rank adapters. Rather than relying on guarantees averaged over multiple random seeds, they studied the concentration of the generalization gap for a single fine-tuning run, yielding sharp, high-probability upper bounds that scale proportionally to $\mathcal{O}(\sqrt{r/N})$ along with a matching theoretical lower bound. Wu et al. [43] also analyzed LoRA generalization but through the lens of Neural Tangent Kernels (NTK). By linearizing the non-linear network dynamics in the wide-model limit, they demonstrated that choosing an adapter rank $r \gtrsim \sqrt{N}$ guarantees the elimination of spurious local minima while maintaining generalization performance comparable to full fine-tuning in underdetermined regimes.

Additionally, the optimization dynamics of factored weight matrices have been heavily studied. Wang et al. [42] explored the implicit bias of LoRA optimization, revealing that the non-convex $A \times B$ factorization intrinsically steers gradient descent toward well-regularized solutions, which acts as a mechanism to mitigate catastrophic forgetting without requiring explicit penalty terms. Tian et al. [35] analyzed convergence properties under different initialization schemes, theoretically demonstrating that standard random initialization often traps LoRA in suboptimal low-rank subspaces; they showed that reaching the optimal minimizers often strictly requires data-driven or “preheated” initializations. Zhu et al. [50] explored the theoretical implications of asymmetric structures in the low-rank factors, proving that the down-projection and up-projection matrices exhibit fundamentally different spectral dynamics and learning speeds during optimization. Finally, Liu et al. [19] provided robust convergence guarantees for LoRA under overparameterized settings, leveraging Polyak-Lojasiewicz (PL) conditions to theoretically guarantee that global minima remain accessible via gradient descent, provided the base network is sufficiently wide.

Despite that variety of approaches tested in previous works, a critical element is missing: these results largely fail to account for the actual spectral structure and energy distribution of the learned adapters. Algebraic rank assumes a maximum entropy scenario where all r components contribute equally or at least expected to be able to contribute equally. Recently, Lion et al. [18] identified this blind spot in their paper introducing PoLAR (Polar-Decomposed Low-Rank Adapter Representation). They demonstrated empirically that standard LoRA suffers from a “directional diversity collapse,” learning updates with stable rank well below the regular algebraic rank of the adapters, often even close to 1. This phenomenon severely degrades the effective utilization of the allocated parameter subspace. Inspired by these practical observations, our work bridges this gap by replacing nominal rank with stable rank directly in the theoretical bounds of expressivity and generalization.

2.3 Stable Rank in Machine Learning

Stable rank has historically emerged as a powerful, continuous tool for analyzing matrix properties in machine learning and random matrix theory, offering distinct advantages over algebraic rank.

Introduced formally by Rudelson and Vershynin [31] in the context of sampling from large matrices, stable rank serves as a continuous measure of effective dimensionality. Tropp [36, 37] expanded heavily on its properties, showing that matrix concentration inequalities in random matrix theory are far more tightly bounded by stable rank than by algebraic dimensions. In the neural network theory, Bartlett et al. [2] and Neyshabur et al. [28] utilized spectral norms and matrix energy distributions to derive generalization bounds

for deep networks, proving that the Lipschitz continuity and spectral complexity of a network better predict generalization than parameter counting.

More recently, stable rank has been also applied to modern transformer architectures. As mentioned, the PoLAR authors [18] documented the tendency of LoRA to collapse into an effective stable rank close to 1. Concurrently, Saada et al. [32] utilized Random Matrix Theory to analyze rank collapse in attention layers, showing that standard softmax attention inherently develops a widening spectral gap between the largest eigenvalue and the rest, leading to rank collapse in width. Furthermore, spectral norm regularizations have been effectively utilized by Miyato et al. [26] and Yoshida et al. [46] to improve generalization by artificially constraining the same metrics that stable rank relies on. These works together suggest that stable rank is not just an alternative metric, but the governing mathematical variable dictating the actual complexity and robustness of the networks, thus it is actually useful for adapters analysis.

2.4 PAC-Bayesian Framework

To properly integrate stable rank into a generalization bound, we require a framework that natively incorporates continuous, energy-based matrix norms rather than combinatorial parameter counting (like VC-dimension). The PAC-Bayesian framework provides exactly this principled approach to deriving data-dependent generalization bounds.

The foundational PAC-Bayes theorem, established by McAllester [24, 23] and extended to neural networks by Shawe-Taylor and Williamson [33], bounds the expected generalization gap of a randomized predictor. Let $\mathcal{P}$ be a prior distribution over hypotheses chosen before seeing the training data, and $\mathcal{Q}$ be a posterior distribution learned after observing data. With probability at least $1 - \delta$ over a dataset of size N , the true expected risk L_0 is bounded by the empirical risk $\hat{L}$ plus a complexity penalty:

$$\mathbb{E}_{w \sim \mathcal{Q}}[L_0(w)] \leq \mathbb{E}_{w \sim \mathcal{Q}}[\hat{L}(w)] + \sqrt{\frac{KL(\mathcal{Q}||\mathcal{P}) + \ln(2N/\delta)}{2N}} \quad (2)$$

where $KL(\mathcal{Q}||\mathcal{P})$ is the Kullback-Leibler divergence between the posterior and the prior.

This framework is exceptionally well-suited for fine-tuning scenarios, including Low-rank adaptation. If we set the prior $\mathcal{P}$ to be an isotropic Gaussian distribution centered on the pre-trained frozen weights W_0 , and the posterior $\mathcal{Q}$ to be a distribution centered on the adapted weights $W_0 + \Delta W$, the KL divergence cleanly reduces to the scaled Frobenius norm of the perturbation: $KL \propto \|\Delta W\|_F^2$.

Building on this, Neyshabur et al. [27] and Bartlett et al. [2] developed spectrally-normalized margin bounds for deep networks by combining the PAC-Bayes theorem with Lipschitz continuity across layers.

More recently, Kratsios et al. [15] established the previous best sample complexity bound of $\tilde{\mathcal{O}}(\sqrt{r/N})$ for asymmetric randomized low-rank adapters. However, rather than utilizing a PAC-Bayesian approach, their analysis relies on constructive approximation theory and empirical Rademacher complexity. Specifically, their method bounds the local Lipschitz regularity of the parameter-to-network map, uses this to estimate the covering numbers of the randomized LoRA function class, and finally applies Dudley’s entropy integral to bound the generalization gap uniformly over the entire hypothesis space. While their approach yields a mathematically sharp sample complexity, we cannot follow this same path. As a result of their framework, the topological dimension of the covering space is intrinsically tied to the ambient parameter count. This lead to an unbreakable proportionality to the algebraic rank, resulting in the $\mathcal{O}(\sqrt{r/N})$ scaling. If we were to introduce a stable rank constraint within their volumetric covering-number approach, it would not effectively reduce the topological dimension of the parameter space, making it theoretically impossible to decouple the final generalization bound from $\sqrt{r}$. In contrast, we adopt the PAC-Bayesian framework because its complexity measure—the Kullback-Leibler divergence between a data-independent prior and the learned posterior—naturally captures the spectral and norm-based properties of the network weights. This allows us to smoothly integrate the stable rank constraint directly into the KL-divergence term, successfully bypassing the ambient algebraic rank entirely to yield a tighter generalization bound that scales with the stable rank rather than r .

3 Mathematical Framework

3.1 Preliminaries

We use standard matrix norm notation: $\|A\|_F = \sqrt{\sum_{i,j} A_{ij}^2} = \sqrt{\sum_i \sigma_i^2(A)}$ denotes the Frobenius norm, and $\|A\|_{op} = \sigma_1(A)$ denotes the spectral (operator) norm. For a matrix $A \in \mathbb{R}^{m \times n}$ with singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_{\min(m,n)} \geq 0$, the stable rank is defined as $\text{sr}(A) = \|A\|_F^2 / \|A\|_{op}^2$. Note that $1 \leq \text{sr}(A) \leq \text{rank}(A) \leq \min(m, n)$.

To provide intuition, if A is a dense matrix sampled from a standard normal distribution $\mathcal{N}(0, 1)$, its stable rank is typically proportional to its algebraic dimension (specifically $\text{sr}(A) \approx \min(m, n)$ for large matrices). Thus, observing a stable rank much closer to 1 in trained weights represents a significant, non-trivial collapse of energy into a few dominant directions, rather than a default mathematical property of all matrices.

Throughout this work for a linear layer with weight matrix W , we consider a LoRA adapter parameterized as $\Delta W = BA$ where $B \in \mathbb{R}^{m \times r}$, $A \in \mathbb{R}^{r \times n}$, and r is the nominal algebraic rank.

3.2 Single Matrix Approximation

We begin by analyzing the fundamental problem of approximating a target matrix A with an estimate $\hat{A}$ under stable rank constraints. This setting isolates the core empirical issue that LoRA faces: even if the nominal rank r is large, the optimization process tends to produce matrices with constrained stable rank, limiting their ability to fit arbitrary targets.

The classical Eckart-Young-Mirsky theorem [6] provides the foundation for understanding low-rank matrix approximation. It states that for any matrix A and integer k , the optimal rank- k approximation in the Frobenius norm is obtained by truncating the Singular Value Decomposition (SVD) to the top k singular values: $\hat{A}_k = \sum_{i=1}^k \sigma_i u_i v_i^T$. However, our setting introduces a crucial distinction. Instead of merely constraining the number of non-zero singular values (algebraic rank), we constrain their distribution through the stable rank $\text{sr}(\hat{A}) = \frac{\sum \sigma_i^2}{\sigma_1^2} \leq S$. Under an algebraic rank constraint, the optimal solution zeros out tail singular values. Under a stable rank constraint, one must actively concentrate the energy, either by shrinking the tail or exploding the top singular value.

Lemma 1. (Interpretable SR-Constrained Approximation Error Lower Bound)

Let A be a target matrix and $\hat{A}$ be an estimate. Let $S_A = \text{sr}(A)$, $S_{\hat{A}} = \text{sr}(\hat{A})$ with $S_{\hat{A}} \leq S_A$, and let $\sigma_1(\hat{A}) = \alpha \sigma_1(A)$ for some $\alpha \geq 0$, where $\sigma_1(\cdot)$ denotes the first (largest) singular value of the corresponding matrix. Then, the Frobenius distance is bounded below by:

$$\|A - \hat{A}\|_F \geq \sigma_1(A) \left(\frac{\sqrt{S_A} - \sqrt{S_{\hat{A}}}}{1 + \sqrt{S_{\hat{A}}}} \right).$$

Доказательство.

By the definition of stable rank, $\text{sr}(A) = \frac{\|A\|_F^2}{\sigma_1(A)^2}$, which implies $\|A\|_F = \sigma_1(A) \sqrt{\text{sr}(A)}$. By the reverse triangle inequality, we have:

$$\|A - \hat{A}\|_F \geq \left| \|A\|_F - \|\hat{A}\|_F \right|.$$

Substituting the stable rank expressions yields:

$$\|A - \hat{A}\|_F \geq |\sigma_1(A) \sqrt{S_A} - \sigma_1(\hat{A}) \sqrt{S_{\hat{A}}}| = \sigma_1(A) \left| \sqrt{S_A} - \alpha \sqrt{S_{\hat{A}}} \right|.$$

We analyze this lower bound by considering two cases for α :

Case 1 ($\alpha \leq 1$):

$$\|A - \hat{A}\|_F \geq \sigma_1(A) \left(\sqrt{S_A} - \sqrt{S_{\hat{A}}} \right).$$

Case 2 ($\alpha > 1$):

$$\|A - \hat{A}\|_F \geq \|A - \hat{A}\|_{op} \geq |\sigma_1(A) - \sigma_1(\hat{A})| = \sigma_1(A)(\alpha - 1).$$

Combining both cases, we can construct a unified lower bound by finding the minimum possible error over all choices of $\alpha \geq 1$:

$$\|A - \hat{A}\|_F \geq \sigma_1(A) \min_{\alpha \geq 1} \max \left(\sqrt{S_A} - \alpha \sqrt{S_{\hat{A}}}, \alpha - 1 \right) \geq \sigma_1(A) \left(\frac{\sqrt{S_A} - \sqrt{S_{\hat{A}}}}{1 + \sqrt{S_{\hat{A}}}} \right).$$

Comparing the bounds from both cases, we note that $\sqrt{S_A} - \sqrt{S_{\hat{A}}} \geq \frac{\sqrt{S_A} - \sqrt{S_{\hat{A}}}}{1 + \sqrt{S_{\hat{A}}}}$. Therefore, the global minimum over α yields:

$$\|A - \hat{A}\|_F \geq \sigma_1(A) \left(\frac{\sqrt{S_A} - \sqrt{S_{\hat{A}}}}{1 + \sqrt{S_{\hat{A}}}} \right).$$

□

This first lemma provides a fundamental lower bound on the approximation accuracy of a high-SR matrix by a low-SR one, establishing the intuition that this bound is primarily dependent on the difference between their stable ranks. This insight naturally leads to the conclusion that when dealing with «difficult» target data that intrinsically requires a high-SR adapter to be predicted accurately, our prediction accuracy will be fundamentally limited if we restrict ourselves to a low-SR adapter.

Lemma 2. (Optimal SR-Constrained Approximation Error Bound)

Assuming the conditions of the previous lemma, let $A_R = \sum_{i=1}^R \sigma_i u_i v_i^T$ be the optimal unconstrained rank- R approximation of A with singular values $\sigma_1 \geq \dots \geq \sigma_R$, and let its stable rank be $\text{sr}(A_R) = \frac{\sum_{i=1}^R \sigma_i^2}{\sigma_1^2}$. For a given maximum stable rank $S \in [1, R]$, let $\mathcal{M}_{R,S}$ be the family of matrices $\hat{A}$ such that $\text{rank}(\hat{A}) \leq R$ and $\text{sr}(\hat{A}) \leq S$. Then the globally optimal approximation $\hat{A} \in \mathcal{M}_{R,S}$ minimizing $\|A - \hat{A}\|_F^2$ is given by $\hat{A} = \sum_{i=1}^R \hat{\sigma}_i u_i v_i^T$. If $\text{sr}(A_R) \leq S$, then $\hat{\sigma}_i = \sigma_i$. If $\text{sr}(A_R) > S$, the optimal singular values are:

$$\hat{\sigma}_1 = \frac{\sigma_1 + \sqrt{S-1} \sqrt{\sum_{j=2}^R \sigma_j^2}}{S}, \quad \hat{\sigma}_i = \gamma \sigma_i \text{ for } i \geq 2,$$

where $\gamma = \frac{\sqrt{S-1} \hat{\sigma}_1}{\sqrt{\sum_{j=2}^R \sigma_j^2}}$. The minimal approximation error is exactly:

$$\|A - \hat{A}\|_F^2 = \sum_{i=R+1}^D \sigma_i^2 + \Delta_S,$$

where the additional penalty Δ_S due to the stable rank constraint is:

$$\Delta_S = \begin{cases} 0 & \text{if } \text{sr}(A_R) \leq S, \\ \frac{1}{S} \left(\sqrt{\sum_{i=2}^R \sigma_i^2} - \sqrt{S-1} \sigma_1 \right)^2 & \text{if } \text{sr}(A_R) > S. \end{cases}$$

Доказательство.

By expanding the squared Frobenius distance, we obtain

$$\|A - \hat{A}\|_F^2 = \|A\|_F^2 + \|\hat{A}\|_F^2 - 2\text{Tr}(A^T \hat{A})$$

The constraints on $\hat{A}$ (rank bounded by R and stable rank bounded by S) depend entirely on its singular values $\hat{\sigma}_i$ and are invariant to its singular vectors. Since $\|A\|_F^2$ is constant and $\|\hat{A}\|_F^2 = \sum_{i=1}^R \hat{\sigma}_i^2$, minimizing the distance for any valid set of singular values requires maximizing the trace $\text{Tr}(A^T \hat{A})$. By von Neumann's trace inequality,

$$\text{Tr}(A^T \hat{A}) \leq \sum_{i=1}^R \sigma_i \hat{\sigma}_i$$

and equality is strictly achieved if and only if the singular vectors of $\hat{A}$ perfectly align with the top R singular vectors of A . Assigning any component of $\hat{A}$ to the tail vectors ($i > R$) of A would pair it with a strictly smaller σ_i , decreasing the trace and increasing the error. Thus, we reduced the problem to finding values $\hat{\sigma}_1 \geq \dots \geq \hat{\sigma}_R \geq 0$ that minimize

$$f(\hat{\sigma}) = \sum_{i=1}^R (\sigma_i - \hat{\sigma}_i)^2$$

subject to the stable rank constraint

$$g(\hat{\sigma}) = \sum_{i=2}^R \hat{\sigma}_i^2 - (S-1)\hat{\sigma}_1^2 \leq 0.$$

This forms a convex optimization problem: the objective function $f(\hat{\sigma})$ is strictly convex (a paraboloid), and the feasible region defined by $g(\hat{\sigma}) \leq 0$ for $\hat{\sigma}_1 > 0$ forms a convex set. Slater's condition is naturally satisfied since there exists an interior point strictly satisfying the inequality e.g., $\hat{\sigma}_1 = 1$, $\hat{\sigma}_{i \geq 2} = 0$ yields $1 - S < 0$ for $S > 1$. Consequently, strong duality holds, and the Karush-Kuhn-Tucker (KKT) conditions are both necessary and sufficient for the global optimum.

If $\text{sr}(A_R) \leq S$, the unconstrained minimum $\hat{\sigma}_i = \sigma_i$ lies inside the feasible region, the Lagrange multiplier λ is zero, and $\Delta_S = 0$. If $\text{sr}(A_R) > S$, the unconstrained minimum lies outside the feasible region. By the complementary slackness condition ($\lambda \cdot g(\hat{\sigma}) = 0$), the optimal solution must lie strictly on the boundary. Thus, the constraint becomes active: $\sum_{i=2}^R \hat{\sigma}_i^2 = (S-1)\hat{\sigma}_1^2$, with $\lambda > 0$.

We define the Lagrangian for the active constraint over the top R components:

$$\mathcal{L} = \frac{1}{2}(\hat{\sigma}_1 - \sigma_1)^2 + \frac{1}{2} \sum_{i=2}^R (\hat{\sigma}_i - \sigma_i)^2 + \frac{\lambda}{2} \left(\sum_{i=2}^R \hat{\sigma}_i^2 - (S-1)\hat{\sigma}_1^2 \right).$$

Setting the partial derivatives with respect to the tail components ($i \geq 2$) to zero yields

$$\hat{\sigma}_i - \sigma_i + \lambda \hat{\sigma}_i = 0 \implies \hat{\sigma}_i = \gamma \sigma_i,$$

where

$$\gamma = \frac{1}{1 + \lambda} > 0.$$

Instead of solving a non-linear system for λ directly, we express the scaling factor γ via $\hat{\sigma}_1$. Substituting $\hat{\sigma}_i = \gamma \sigma_i$ into the active constraint gives

$$\gamma^2 \sum_{i=2}^R \sigma_i^2 = (S-1)\hat{\sigma}_1^2.$$

Letting $\|v\| = \sqrt{\sum_{i=2}^R \sigma_i^2}$, we obtain

$$\gamma = \frac{\sqrt{S-1}\hat{\sigma}_1}{\|v\|}.$$

Substituting this exact dependency back into the original objective reduces the problem to a 1D unconstrained minimization over $\hat{\sigma}_1$:

$$f(\hat{\sigma}_1) = (\hat{\sigma}_1 - \sigma_1)^2 + \sum_{i=2}^R (\gamma \sigma_i - \sigma_i)^2 = (\hat{\sigma}_1 - \sigma_1)^2 + (\|v\| - \sqrt{S-1}\hat{\sigma}_1)^2.$$

Setting $f'(\hat{\sigma}_1) = 0$ yields:

$$\hat{\sigma}_1 - \sigma_1 - \sqrt{S-1}(\|v\| - \sqrt{S-1}\hat{\sigma}_1) = 0 \implies \hat{\sigma}_1 = \frac{\sigma_1 + \sqrt{S-1}\|v\|}{S}.$$

Since $\text{sr}(A_R) > S$, we have $\|v\|^2 > (S-1)\sigma_1^2$, which strictly ensures $\hat{\sigma}_1 > \sigma_1$. This uncovers a fundamental property: to optimally enforce a lowered stable rank, one cannot merely shrink the tail components; the

principal component must simultaneously be inflated. Furthermore, this guarantees that singular values remain properly ordered, as

$$\hat{\sigma}_2 = \gamma \sigma_2 = \frac{\sqrt{S-1}\sigma_2}{\|v\|} \hat{\sigma}_1 < \hat{\sigma}_1.$$

Finally, plugging the optimal $\hat{\sigma}_1$ back into the objective gives the exact minimal squared error Δ_S introduced on the top R components:

$$\Delta_S = \left(\frac{\sqrt{S-1}\|v\| - (S-1)\sigma_1}{S} \right)^2 + \left(\frac{\|v\| - \sqrt{S-1}\sigma_1}{S} \right)^2 = \frac{1}{S}(\|v\| - \sqrt{S-1}\sigma_1)^2.$$

Adding the irreducible tail error $\sum_{i=R+1}^D \sigma_i^2$ provides the absolute minimal distance over the entire $\mathcal{M}_{R,S}$ family. $\square$

Consequently, the second lemma does not merely establish an upper bound, but actively constructs the theoretically optimal approximation matrix. It provides the exact analytical formulation for the minimal error introduced when a low stable rank constraint is enforced along with a regular rank

3.3 Fully Connected Network Approximation

Having established the core matrix approximation mechanics, we can now evaluate the expressive power of LoRA within a multi-layer deep network. Zeng and Lee [48] previously proved that rank- r LoRA adapters can represent any rank- r perturbation of pretrained weights across layers, establishing an existence proof for algebraic expressivity. However, their bound implies perfect capacity up to the nominal rank.

By extending their, we show that following the same compositional logic but imposing the empirically observed stable rank limits yields a significantly more restricted bound. Theorem 1 demonstrates that the approximation error at each layer is explicitly dictated by the stable rank penalty Δ_S derived in the lemma 2. Furthermore, because the network is deep, these layer-wise constraints do not simply sum; they are multiplicatively amplified by the spectral norms of subsequent layers, meaning an inability to model high-SR features in early layers exponentially degrades the output expressivity.

3.3.1 Notation

Let $FNN_{L,D}(\cdot; (W_l)_{l=1}^L, (b_l)_{l=1}^L)$ be a fully connected neural network with L layers of width D and ReLU activation functions. $W_l \in \mathbb{R}^{D \times D}$ are the weight matrices and $b_l \in \mathbb{R}^D$ are the bias vectors.

In this work we use:

$$\begin{aligned} \text{target FNN} \quad \bar{f} &:= FNN_{\bar{L},D}(\cdot; (\bar{W}_l)_{l=1}^{\bar{L}}, (\bar{b}_l)_{l=1}^{\bar{L}}), \\ \text{frozen FNN} \quad f_0 &:= FNN_{L,D}(\cdot; (W_l)_{l=1}^L, (b_l)_{l=1}^L), \\ \text{adapted FNN} \quad f &:= FNN_{L,D}(\cdot; (W_l + \Delta W_l)_{l=1}^L, (\hat{b}_l)_{l=1}^L) \end{aligned}$$

where $b_l, \bar{b}_l, \hat{b}_l \in \mathbb{R}^D$, $W_l, \bar{W}_l, \Delta W_l \in \mathbb{R}^{D \times D}$, and $\Delta W_l = A_l B_l$, where $A_l \in \mathbb{R}^{D \times R}$ and $B_l \in \mathbb{R}^{R \times D}$ and $R \leq D$ is the nominal LoRA rank. Additionally, we assume $\bar{L} = L$.

3.3.2 Expressive power theorem

Building upon the approximation framework and applying the matrix reconstruction residual from Lemma 2, we derive the following expressivity bound:

Theorem 1.

Let $x \in \mathbb{R}^d$ be the random input with covariance matrix $\Sigma = \mathbb{E}[xx^T]$. Let $E_l = \bar{W}_l - W_l$ be the required weight discrepancy at layer l to match the target. Let $\sigma_{l,i}$ be the singular values of E_l , and let $E_{l,R}$ be its optimal unconstrained rank- R approximation from Eckart-Young-Mirsky theorem. Assume $\hat{b}_l = \bar{b}_l$.

Then there exist adapters ΔW_l of rank $\leq R$ and $\forall l \in \overline{1, \bar{L}}$ $\text{sr}(\Delta W_l) \leq S$ such that the expected approximation error is bounded by:

$$\mathbb{E}\|f(\mathbf{x}) - \bar{f}(\mathbf{x})\|_2 \leq \beta \sum_{l=1}^L \left(\prod_{k=l+1}^L \|\bar{W}_k\|_2 \right) \tau_l$$

Where β accounts for the magnitude of the parameters and the input:

$$\beta = \max_{i \in [L]} \left(\underbrace{\left(\prod_{j=1}^i \|\bar{W}_j\| \right)}_{\text{Input Term}} \sqrt{\text{tr}(\Sigma)} + \underbrace{\sum_{j=1}^i \left(\prod_{k=j+1}^i \|\bar{W}_k\| \right) \|\bar{b}_j\|}_{\text{Bias Term}} \right)$$

and $\mathcal{T}_l$ is the localized error derived from Lemma 2:

$$\mathcal{T}_l = \sqrt{\sum_{j=R+1}^D \sigma_{l,j}^2 + \Delta_{S,l}}$$

with the stable rank penalty $\Delta_{S,l}$ defined as:

$$\Delta_{S,l} = \begin{cases} 0 & \text{if } \text{sr}(E_{l,R}) \leq S \\ \frac{1}{S} \left(\sqrt{\sum_{j=2}^R \sigma_{l,j}^2} - \sqrt{S-1} \sigma_{l,1} \right)^2 & \text{if } \text{sr}(E_{l,R}) > S \end{cases}$$

Доказательство.

1. Linearization: Let h_l be the output of layer l in the adapted model, and $\bar{h}_l$ in the target model.

$$\begin{aligned} h_l &= \mathbf{ReLU}((W_l + \Delta W_l)h_{l-1} + \bar{b}_l) \\ \bar{h}_l &= \mathbf{ReLU}(\bar{W}_l \bar{h}_{l-1} + \bar{b}_l) \end{aligned}$$

We define the error vector

$$e_l = h_l - \bar{h}_l = \mathbf{ReLU}((W_l + \Delta W_l)h_{l-1} + \bar{b}_l) - \mathbf{ReLU}(\bar{W}_l \bar{h}_{l-1} + \bar{b}_l)$$

As $\mathbf{ReLU}$ is 1-Lipschitz

$$\|e_l\|_2 \leq \|((W_l + \Delta W_l)h_{l-1} + \bar{b}_l) - (\bar{W}_l \bar{h}_{l-1} + \bar{b}_l)\|_2$$

Add and subtract $\bar{W}_l h_{l-1}$:

$$\|e_l\|_2 \leq \|(W_l + \Delta W_l - \bar{W}_l)h_{l-1} + \bar{W}_l(h_{l-1} - \bar{h}_{l-1})\|_2$$

Let $M_l = W_l + \Delta W_l - \bar{W}_l$ be the approximation residual matrix for layer l .

$$\|e_l\|_2 \leq \|M_l h_{l-1} + \bar{W}_l e_{l-1}\|_2$$

2. Unrolling the recursion: By induction from $l = 1$ to L (with $e_0 = 0$):

$$\|e_L\|_2 \leq \sum_{l=1}^L \left\| \prod_{k=l+1}^L \bar{W}_k \right\|_2 \|M_l\|_2 \|h_{l-1}\|_2$$

3. Constructing the Adapter ΔW_l : We need to minimize $\|M_l\|_F = \|(W_l + \Delta W_l) - \bar{W}_l\|_F$. This is equivalent to approximating $E_l = \bar{W}_l - W_l$ with ΔW_l . Thus we can apply Lemma 2 and this yields the Frobenius error:

$$\|M_l\|_F = \mathcal{T}_l = \sqrt{\sum_{j=R+1}^D \sigma_{l,j}^2 + \Delta_{S,l}}$$

Since $\|M_l\|_2 \leq \|M_l\|_F$, we use $\mathcal{T}_l$ as the bound for the matrix norm.

4. Handling the intermediate norms: We look at $\mathbb{E}\|e_L\|_2$. The term depends on $\mathbb{E}\|h_{l-1}\|_2$. $\|h_{l-1}\|_2 \leq (\prod_{j=1}^{l-1} \|\bar{W}_j\|_2) \|\mathbf{x}\|_2$. For the expectation, $\mathbb{E}\|\mathbf{x}\|_2 \leq \sqrt{\mathbb{E}\|\mathbf{x}\|^2} = \sqrt{\text{tr}(\Sigma)}$.

We want to bound $\|\bar{h}_i\|_2$.

$$\begin{aligned}\|\bar{h}_1\| &\leq \|\bar{W}_1\| \|\mathbf{x}\| + \|\bar{b}_1\| \\ \|\bar{h}_2\| &\leq \|\bar{W}_2\| \|\bar{h}_1\| + \|\bar{b}_2\|\end{aligned}$$

Substitute the bound for Layer 1:

$$\begin{aligned}\|\bar{h}_2\| &\leq \|\bar{W}_2\| (\|\bar{W}_1\| \|\mathbf{x}\| + \|\bar{b}_1\|) + \|\bar{b}_2\| \\ \|\bar{h}_2\| &\leq \underbrace{\|\bar{W}_2\| \|\bar{W}_1\| \|\mathbf{x}\|}_{\text{Input Contribution}} + \underbrace{\|\bar{W}_2\| \|\bar{b}_1\| + \|\bar{b}_2\|}_{\text{Accumulated Bias Contribution}}\end{aligned}$$

By induction, the magnitude of the activation at layer i is bounded by:

$$\|\bar{h}_i\| \leq \left(\prod_{j=1}^i \|\bar{W}_j\| \right) \|\mathbf{x}\| + \sum_{j=1}^i \left(\prod_{k=j+1}^i \|\bar{W}_k\| \right) \|\bar{b}_j\|$$

Taking the expectation and the maximum over layers we get the β defined in theorem conditions

5. Gather up the terms:

$$\mathbb{E} \|f(\mathbf{x}) - \bar{f}(\mathbf{x})\|_2 = \mathbb{E} \|e_L\|_2 \leq \beta \sum_{l=1}^L \left(\prod_{k=l+1}^L \|\bar{W}_k\|_2 \right) \mathcal{T}_l$$

□

While this theorem relies on sequential worst-case norm accumulations, rendering the numerical bound loose in practical highly-dimensional settings, it successfully isolates the exact mechanic by which stable rank limits expressiveness. It reveals that the actual capacity of an adapter is not dictated by whether its nominal rank is large enough to contain the target matrix, as previous bounds imply, but by whether the optimizer can sufficiently distribute the energy across its singular values to capture the target's effective dimension. When the optimization process is intrinsically biased toward low-SR matrices, $\mathcal{T}_l$ explodes for high-SR targets.

3.4 Transformer Network Approximation

Extending our analysis from FNNs to Transformers introduces several layers of additional complexity that require careful mathematical treatment. Unlike FNNs with fixed affine transformations, the Transformer's attention mechanism computes weights dynamically based on the input through the softmax of Query-Key scores. This means the effective network transformation depends non-linearly on the input distribution. Furthermore, each transformer block possesses dual error propagation paths via the residual connections surrounding both the Multi-Head Attention (MHA) and FFN sub-layers, causing errors at layer l to accumulate from both sources.

The sensitivity of the softmax function itself also amplifies parameter errors, a dynamic we encapsulate through a score sensitivity constant $C_{score,l}^h$. By carefully decomposing the local distortion $\mathcal{D}_l$ into FFN parameter errors and attention head distortions, we can establish how stable rank limits on the individual Q, K, V , and O projection matrices affects the Transformer's total expressive capacity.

3.4.1 Notation

Let

$$TN_{L,D}(\cdot; ((W_{Ql}^h, W_{Kl}^h, W_{Vl}^h, W_{Ol}^h)_{h=1}^H, (W_{1l}, W_{2l}, b_{1l}, b_{2l}))_{l=1}^L, W_o)$$

be a transformer network with L transformer blocks and an output linear layer parameterized by W_o . For theoretical feasibility we assume no dropout or normalisation operations. Each block is a composition of an H -head self-attention mechanism [39] with weights $(W_{Ql}^h, W_{Kl}^h, W_{Vl}^h, W_{Ol}^h)_{h=1}^H$ and a 2-layer FFN with weights (W_{1l}, W_{2l}) and biases (b_{1l}, b_{2l}) . Additionally, we assume all $W \in \mathbb{R}^{D \times D}$, $b \in \mathbb{R}^D$. We use the same prefix notation as before:

$$\begin{aligned}\text{target TN} \quad \bar{f} &:= TN_{L,D}(\cdot; ((\bar{W}_{Ql}^h, \bar{W}_{Kl}^h, \bar{W}_{Vl}^h, \bar{W}_{Ol}^h)_{h=1}^H, (\bar{W}_{1l}, \bar{W}_{2l}, \bar{b}_{1l}, \bar{b}_{2l}))_{l=1}^L, \bar{W}_o), \\ \text{frozen TN} \quad f_0 &:= TN_{L,D}(\cdot; ((W_{Ql}^h, W_{Kl}^h, W_{Vl}^h, W_{Ol}^h)_{h=1}^H, (W_{1l}, W_{2l}, b_{1l}, b_{2l}))_{l=1}^L, W_o),\end{aligned}$$

$$\begin{aligned} \text{adapted TN } f &:= TN_{L,D}(\cdot; ((W_{Ql}^h + \Delta W_{Ql}^h, W_{Kl}^h + \Delta W_{Kl}^h, W_{Vl}^h + \Delta W_{Vl}^h, W_{Ol}^h + \Delta W_{Ol}^h)_{h=1}^H, \\ &\quad (W_{1l} + \Delta W_{1l}, W_{2l} + \Delta W_{2l}, \widehat{b_{1l}}, \widehat{b_{2l}}))_{l=1}^L, W_o + \Delta W_o) \end{aligned}$$

For a matrix $G \in \mathbb{R}^{D \times D}$, let its optimal rank- R approximation be G_R with singular values $\sigma_1(G) \geq \dots \geq \sigma_R(G)$. Let $\mathcal{T}(G)$ be the localized SR error from Lemma 2:

$$\mathcal{T}(G) = \sqrt{\sum_{i=R+1}^D \sigma_i^2(G) + \Delta_S(G)}$$

where $\Delta_S(G) = 0$ if $\text{sr}(G_R) \leq S$, and $\Delta_S(G) = \frac{1}{S} \left(\sqrt{\sum_{i=2}^R \sigma_i^2(G)} - \sqrt{S-1} \sigma_1(G) \right)^2$ if $\text{sr}(G_R) > S$.

3.4.2 Expressive power of transformer network

Following the block decomposition approach used in [47] and extending the rigorous Lipschitz analysis from [41], we formulate the following bound tracking the accumulation of stable rank constraints in a self-attention architecture:

Theorem 2.

Let $x \in \mathbb{R}^{D \times N}$ be the random input sequence. Assume $\overline{b_{..}} = \widehat{b_{..}}$. Let $z_l \in \mathbb{R}^{D \times N}$ be the output of l block in the adapted model and $\overline{z}_l$ be the output of the same block in the target model. Consider $z_0 = \overline{z}_0 = x$. Let us define per matrix discrepancies as follows:

$$\begin{aligned} \forall l \in \overline{1, L}, h \in \overline{1, H}, \forall \text{role} \in \{Q, K, V, O\} : \quad \mathcal{T}_{\text{role}, l}^h &= \mathcal{T}(W_{\text{role}, l}^h) = \|W_{\text{role}, l}^h - \overline{W_{\text{role}, l}^h}\|_F \\ \forall l \in \overline{1, L}, \forall \text{role} \in \{1, 2\} : \quad \mathcal{T}_{\text{role}, l} &= \mathcal{T}(W_{\text{role}, l}) = \|W_{\text{role}, l}^h - \overline{W_{\text{role}, l}^h}\|_F \\ \mathcal{T}_o &= \mathcal{T}(W_o) = \|W_o - \overline{W_o}\|_F \end{aligned}$$

Additionally we define the following bound constants:

$$\begin{aligned} \forall l \in \overline{1, L}, h \in \overline{1, H}, \forall \text{role} \in \{Q, K, V, O\} : \quad \mu_{\text{role}, l}^h &= \|\overline{W_{\text{role}, l}^h}\|_2 + \mathcal{T}_{\text{role}, l}^h \\ \forall l \in \overline{1, L}, \forall \text{role} \in \{1, 2\} : \quad \mu_{\text{role}, l} &= \|\overline{W_{\text{role}, l}^h}\|_2 + \mathcal{T}_{\text{role}, l} \\ \mu_o &= \|\overline{W_o}\|_2 + \mathcal{T}_o \\ \beta_0 &= \sqrt{\mathbb{E}[\|x\|_F^2]}, \quad \text{for } l > 0: \beta_l = \beta_{l-1} \cdot (1 + \mu_{2, l} \mu_{1, l}) \left(1 + \sum_{h=1}^H \mu_{O, l}^h \mu_{V, l}^h \right) \\ \beta &= \beta_L \end{aligned}$$

Then there exists a $\leq S$ stable rank LoRA adapter configuration such that

$$\mathbb{E} \|f(x) - \overline{f}(x)\|_F \leq \beta \sum_{l=1}^L \left(\prod_{k=l+1}^L \gamma_k \right) \mathcal{D}_l$$

where:

1. γ_k is the Lipschitz constant of target layer k with respect to the input:

$$\begin{aligned} \gamma_k &= \|\overline{W_{2, k}^h}\| \|\overline{W_{1, k}^h}\| \left(1 + \sum_{h=1}^H \Psi_k^h \right) \\ \Psi_k^h &= \|\overline{W_{O, k}^h}\|_2 \left(\|\overline{W_{V, k}^h}\|_2 + \frac{\lambda_{sm} \beta^2}{\sqrt{D}} \|\overline{W_{V, k}^h}\|_2 \left(\mu_{Q, k}^h \|\overline{W_{K, k}^h}\|_2 + \|\overline{W_{K, k}^h}\|_2 \|\overline{W_{Q, k}^h}\|_2 \right) \right) \end{aligned}$$

2. $\mathcal{D}_l$ is the Local Distortion at layer l , defined as:

$$\mathcal{D}_l = \mathcal{T}_{2,l} + \|\overline{W_{2,l}}\|_2 \mathcal{T}_{1,l} + \|\overline{W_{2,l}}\|_2 \|\overline{W_{1,l}}\|_2 \sum_{h=1}^H \mathcal{D}_{att,l}^h$$

With the per-head attention distortion $\mathcal{D}_{att,l}^h$:

$$\mathcal{D}_{att,l}^h = \mu_{V,l}^h \mathcal{T}_{O,l}^h + \|\overline{W_{O,l}^h}\|_2 \mathcal{T}_{V,l}^h + C_{score,l}^h (\mu_{Q,l}^h \mathcal{T}_{K,l}^h + \|\overline{W_{K,l}^h}\|_2 \mathcal{T}_{Q,l}^h)$$

And the score sensitivity constant:

$$C_{score,l}^h = \frac{\lambda_{sm}}{\sqrt{D}} \|\overline{W_{O,l}^h}\|_2 \|\overline{W_{V,l}^h}\|_2 \beta$$

Доказательство.

Let $E_l = z_l - \bar{z}_l$ be the error at the output of layer l .

The output of layer l is $z_l = \text{FFN}_l(z_{l-1} + \text{MHA}_l(z_{l-1}))$. Let $Y_l = z_{l-1} + \text{MHA}_l(z_{l-1})$ be the intermediate state.

$$\begin{aligned} E_l &= \text{FFN}_l(Y_l) - \overline{\text{FFN}_l(Y_l)} \\ \|E_l\|_F &\leq \underbrace{\|\text{FFN}_l(Y_l) - \overline{\text{FFN}_l(Y_l)}\|_F}_{\text{FFN Parameter Error}} + \underbrace{\|\overline{\text{FFN}_l(Y_l)} - \overline{\text{FFN}_l(\bar{Y}_l)}\|_F}_{\text{FFN Input Error}} \end{aligned}$$

The FFN part of TN layer l is $W_{2,l} \text{ReLU}(W_{1,l} x)$. Then the FFN parameter error:

$$\begin{aligned} \|W_{2,l} \text{ReLU}(W_{1,l} Y_l) - \overline{W_{2,l}} \text{ReLU}(\overline{W_{1,l}} Y_l)\|_F &\leq \|(W_{2,l} - \overline{W_{2,l}}) \text{ReLU}(W_{1,l} Y_l)\|_F + \|\overline{W_{2,l}}\|_2 \|\text{ReLU}(W_{1,l} Y_l) - \text{ReLU}(\overline{W_{1,l}} Y_l)\|_F \\ &\leq \mathcal{T}_{2,l} \|Y_l\|_F + \|\overline{W_{2,l}}\|_2 \cdot \|(W_{1,l} - \overline{W_{1,l}}) Y_l\|_F \leq \beta \mathcal{T}_{2,l} + \|\overline{W_{2,l}}\|_2 \beta \mathcal{T}_{1,l} \end{aligned}$$

The FFN Input Error is bounded by the following

$$\|\overline{W_{2,l}} (\text{ReLU}(\overline{W_{1,l}} Y_l) - \text{ReLU}(\overline{W_{1,l}} \bar{Y}_l))\|_F \leq \|\overline{W_{2,l}}\|_2 \cdot \|\overline{W_{1,l}}\|_2 \cdot \|Y_l - \bar{Y}_l\|_F$$

Therefore the combined total FFN error bound:

$$\|E_l\|_F \leq \beta (\mathcal{T}_{2,l} + \|\overline{W_{2,l}}\|_2 \mathcal{T}_{1,l}) + \|\overline{W_{2,l}}\|_2 \cdot \|\overline{W_{1,l}}\|_2 \cdot \|Y_l - \bar{Y}_l\|_F$$

Now we analyze the Attention error $\|Y_l - \bar{Y}_l\|_F$. Since $Y_l = z_{l-1} + \sum_h \text{Head}_h^l$, and $E_{l-1} = z_{l-1} - \bar{z}_{l-1}$:

$$\|Y_l - \bar{Y}_l\|_F \leq \|E_{l-1}\|_F + \sum_{h=1}^H \|\text{Head}_h^l(z_{l-1}) - \overline{\text{Head}_h^l(z_{l-1})}\|_F$$

A single head output:

$$\text{Head}_h^l(z_{l-1}) = W_{O,l}^h V_l^h (A_l^h)^T$$

where

$$V_l^h = W_{V,l}^h z_{l-1}, \quad A_l^h = \text{Softmax}(S_l^h), \quad S_l^h = \frac{1}{\sqrt{D}} (K_l^h)^T Q_l^h, \quad K_l^h = W_{K,l}^h z_{l-1}, \quad Q_l^h = W_{Q,l}^h z_{l-1}$$

As with the FFN part, we decompose the error $\|\text{Head} - \overline{\text{Head}}\|_F$ into Head parameter Error (functions of $\mathcal{T}$) and Head Input Error (functions of E_{l-1}).

Output projection $W_{O,l}^h$:

$$W_{O,l}^h V_l^h (A_l^h)^T - \overline{W_{O,l}^h} V_l^h (\bar{A}_l^h)^T = (W_{O,l}^h - \overline{W_{O,l}^h}) V_l^h (A_l^h)^T + \overline{W_{O,l}^h} \left(V_l^h (A_l^h)^T - \bar{V}_l^h (\bar{A}_l^h)^T \right)$$

For the first term the following holds:

$$\|(W_{O,l}^h - \overline{W_{O,l}^h}) V_l^h (A_l^h)^T\|_F \leq \|W_{O,l}^h - \overline{W_{O,l}^h}\|_F \cdot \|V_l^h (A_l^h)^T\|_2 \leq \|W_{O,l}^h - \overline{W_{O,l}^h}\|_F \cdot \|V_l^h (A_l^h)^T\|_F \leq$$

$$\leq \mathcal{T}_{O,l}^h \cdot \|V_l^h\|_F \cdot \|(A_l^h)^T\|_2 \leq \mathcal{T}_{O,l}^h \cdot \|W_{V,l}^h\|_F \cdot 1 \leq \mathcal{T}_{O,l}^h \cdot \|W_{V,l}^h\|_2 \cdot \|z_{l-1}\|_F \leq \mathcal{T}_{O,l}^h \cdot \mu_{V,l}^h \cdot \beta$$

For the second term:

$$\begin{aligned} \left\| \overline{W_{O,l}^h} \left(V_l^h (A_l^h)^T - \overline{V_l^h} (\overline{A_l^h})^T \right) \right\| &\leq \left\| \overline{W_{O,l}^h} \right\|_2 \left\| V_l^h (A_l^h)^T - \overline{V_l^h} (\overline{A_l^h})^T \right\|_F \\ V_l^h (A_l^h)^T - \overline{V_l^h} (\overline{A_l^h})^T &= (V_l^h - \overline{V_l^h}) (A_l^h)^T + \overline{V_l^h} (A_l^h - (\overline{A_l^h})^T) \end{aligned}$$

V matrices difference:

$$\begin{aligned} V_l^h - \overline{V_l^h} &= (W_{V,l}^h - \overline{W_{V,l}^h}) z_{l-1} + \overline{W_{V,l}^h} (z_{l-1} - \overline{z_{l-1}}) \\ \left\| V_l^h - \overline{V_l^h} \right\|_F &\leq \left\| (W_{V,l}^h - \overline{W_{V,l}^h}) z_{l-1} \right\|_F + \left\| \overline{W_{V,l}^h} (z_{l-1} - \overline{z_{l-1}}) \right\|_F \leq \mathcal{T}_{V,l}^h \beta + \left\| \overline{W_{V,l}^h} \right\|_2 \cdot \|E_{l-1}\|_F \end{aligned}$$

A matrices difference:

$$\left\| A_l^h - \overline{A_l^h} \right\|_F \leq \lambda_{sm} \left\| S_l^h - \overline{S_l^h} \right\|_F$$

Additionally $\left\| \overline{V_l^h} \right\|_F \leq \left\| \overline{W_{V,l}^h} \right\|_2 \cdot \beta$, thus:

$$\left\| \overline{V_l^h} (A_l^h - (\overline{A_l^h})^T) \right\| \leq \lambda_{sm} \cdot \beta \cdot \left\| \overline{W_{V,l}^h} \right\|_2 \cdot \left\| S_l^h - \overline{S_l^h} \right\|_F$$

Now we need to bound the score matrices S difference:

$$S_l^h - \overline{S_l^h} = \frac{1}{\sqrt{D}} ((K_l^h)^T Q_l^h - (\overline{K_l^h})^T \overline{Q_l^h}) = \frac{1}{\sqrt{D}} \left((K_l^h - \overline{K_l^h})^T Q_l^h + (\overline{K_l^h})^T (Q_l^h - \overline{Q_l^h}) \right)$$

Q matrices difference:

$$\begin{aligned} Q_l^h - \overline{Q_l^h} &= (W_{Q,l}^h - \overline{W_{Q,l}^h}) z_{l-1} + \overline{W_{Q,l}^h} (z_{l-1} - \overline{z_{l-1}}) \\ \left\| Q_l^h - \overline{Q_l^h} \right\|_F &\leq \mathcal{T}_{Q,l}^h \cdot \beta + \left\| \overline{W_{Q,l}^h} \right\|_2 \cdot \|E_{l-1}\|_F \end{aligned}$$

K matrices difference:

$$\begin{aligned} K_l^h - \overline{K_l^h} &= (W_{K,l}^h - \overline{W_{K,l}^h}) z_{l-1} + \overline{W_{K,l}^h} (z_{l-1} - \overline{z_{l-1}}) \\ \left\| K_l^h - \overline{K_l^h} \right\|_F &\leq \mathcal{T}_{K,l}^h \cdot \beta + \left\| \overline{W_{K,l}^h} \right\|_2 \cdot \|E_{l-1}\|_F \end{aligned}$$

Now we substitute back step by step:

$$\begin{aligned} \left\| S_l^h - \overline{S_l^h} \right\|_F &\leq \frac{1}{\sqrt{D}} \left(\left\| (K_l^h - \overline{K_l^h})^T Q_l^h \right\|_F + \left\| (\overline{K_l^h})^T (Q_l^h - \overline{Q_l^h}) \right\|_F \right) \leq \\ &\leq \frac{1}{\sqrt{D}} \left(\left\| K_l^h - \overline{K_l^h} \right\|_F \|W_{Q,l}^h\|_2 \|z_{l-1}\|_F + \left\| \overline{W_{K,l}^h} \right\|_2 \|\overline{z_{l-1}}\|_F \left\| Q_l^h - \overline{Q_l^h} \right\|_F \right) \leq \\ &\leq \frac{\beta}{\sqrt{D}} \left(\left\| K_l^h - \overline{K_l^h} \right\|_F \mu_{Q,l}^h + \left\| \overline{W_{K,l}^h} \right\|_2 \left\| Q_l^h - \overline{Q_l^h} \right\|_F \right) \leq \\ &\leq \frac{\beta}{\sqrt{D}} \left(\mu_{Q,l}^h \mathcal{T}_{K,l}^h \beta + \left\| \overline{W_{K,l}^h} \right\|_2 \mathcal{T}_{Q,l}^h \beta \right) + \frac{\beta}{\sqrt{D}} \left(\mu_{Q,l}^h \left\| \overline{W_{K,l}^h} \right\|_2 + \left\| \overline{W_{K,l}^h} \right\|_2 \left\| \overline{W_{Q,l}^h} \right\|_2 \right) \|E_{l-1}\|_F \\ &\quad \left\| \overline{V_l^h} (A_l^h - (\overline{A_l^h})^T) \right\|_F \leq \\ &\leq \lambda_{sm} \beta \left\| \overline{W_{V,l}^h} \right\|_2 \left[\frac{\beta}{\sqrt{D}} \left(\mu_{Q,l}^h \mathcal{T}_{K,l}^h \beta + \left\| \overline{W_{K,l}^h} \right\|_2 \mathcal{T}_{Q,l}^h \beta \right) + \frac{\beta}{\sqrt{D}} \left(\mu_{Q,l}^h \left\| \overline{W_{K,l}^h} \right\|_2 + \left\| \overline{W_{K,l}^h} \right\|_2 \left\| \overline{W_{Q,l}^h} \right\|_2 \right) \|E_{l-1}\|_F \right] \\ &\quad \left\| V_l^h (A_l^h)^T - \overline{V_l^h} (\overline{A_l^h})^T \right\|_F \leq \left\| V_l^h - \overline{V_l^h} \right\|_F \left\| (A_l^h)^T \right\|_2 + \left\| \overline{V_l^h} (A_l^h - (\overline{A_l^h})^T) \right\|_F \leq \\ &\leq \mathcal{T}_{V,l}^h \beta + \left\| \overline{W_{V,l}^h} \right\|_2 \|E_{l-1}\|_F + \frac{\lambda_{sm} \beta^2}{\sqrt{D}} \left\| \overline{W_{V,l}^h} \right\|_2 \left(\mu_{Q,l}^h \mathcal{T}_{K,l}^h \beta + \left\| \overline{W_{K,l}^h} \right\|_2 \mathcal{T}_{Q,l}^h \beta \right) + \end{aligned}$$

$$+ \frac{\lambda_{sm}\beta^2}{\sqrt{D}} \left\| \overline{W_{V,l}^h} \right\|_2 \left(\mu_{Q,l}^h \left\| \overline{W_{K,l}^h} \right\|_2 + \left\| \overline{W_{K,l}^h} \right\|_2 \left\| \overline{W_{Q,l}^h} \right\|_2 \right) \|E_{l-1}\|_F$$

Getting back to the head error:

$$\begin{aligned} \left\| \text{Head}_h^l(z_{l-1}) - \overline{\text{Head}_h^l(\overline{z_{l-1}})} \right\|_F &\leq \mathcal{T}_{O,l}^h \mu_{V,l}^h \beta + \left\| \overline{W_{O,l}^h} \right\|_2 \left\| V_l^h (A_l^h)^T - \overline{V_l^h} (\overline{A_l^h})^T \right\|_F \leq \\ &\leq \mathcal{T}_{O,l}^h \mu_{V,l}^h \beta + \left\| \overline{W_{O,l}^h} \right\|_2 \left[\mathcal{T}_{V,l}^h \beta + \frac{\lambda_{sm}\beta^2}{\sqrt{D}} \left\| \overline{W_{V,l}^h} \right\|_2 \left(\mu_{Q,l}^h \mathcal{T}_{K,l}^h \beta + \left\| \overline{W_{K,l}^h} \right\|_2 \mathcal{T}_{Q,l}^h \beta \right) \right] + \\ &+ \left\| \overline{W_{O,l}^h} \right\|_2 \left[\left\| \overline{W_{V,l}^h} \right\|_2 + \frac{\lambda_{sm}\beta^2}{\sqrt{D}} \left\| \overline{W_{V,l}^h} \right\|_2 \left(\mu_{Q,l}^h \left\| \overline{W_{K,l}^h} \right\|_2 + \left\| \overline{W_{K,l}^h} \right\|_2 \left\| \overline{W_{Q,l}^h} \right\|_2 \right) \right] \|E_{l-1}\|_F \end{aligned}$$

Grouping the error into Head parameter Error ($\mathcal{D}_{att,l}^h$) and Head Input Error (Ψ_l^h):

$$\left\| \text{Head}_h^l(z_{l-1}) - \overline{\text{Head}_h^l(\overline{z_{l-1}})} \right\|_F \leq \beta \mathcal{D}_{att,l}^h + \Psi_l^h \|E_{l-1}\|_F$$

where

$$\begin{aligned} \mathcal{D}_{att,l}^h &= \mathcal{T}_{O,l}^h \mu_{V,l}^h \beta + \left\| \overline{W_{O,l}^h} \right\|_2 \mathcal{T}_{V,l}^h + \frac{\lambda_{sm}\beta}{\sqrt{D}} \left\| \overline{W_{O,l}^h} \right\|_2 \left\| \overline{W_{V,l}^h} \right\|_2 \left(\mu_{Q,l}^h \mathcal{T}_{K,l}^h \beta + \left\| \overline{W_{K,l}^h} \right\|_2 \mathcal{T}_{Q,l}^h \beta \right) \\ \Psi_l^h &= \left\| \overline{W_{O,l}^h} \right\|_2 \left(\left\| \overline{W_{V,l}^h} \right\|_2 + \frac{\lambda_{sm}\beta^2}{\sqrt{D}} \left\| \overline{W_{V,l}^h} \right\|_2 \left(\mu_{Q,l}^h \left\| \overline{W_{K,l}^h} \right\|_2 + \left\| \overline{W_{K,l}^h} \right\|_2 \left\| \overline{W_{Q,l}^h} \right\|_2 \right) \right) \end{aligned}$$

The output of the multi-head attention block combined with the residual connection is denoted as Y_l :

$$Y_l = z_{l-1} + \sum_{h=1}^H \text{Head}_h^l(z_{l-1})$$

The difference before the FFN layer is bounded by summing the individual head errors:

$$\begin{aligned} \|Y_l - \overline{Y_l}\|_F &= \left\| z_{l-1} - \overline{z_{l-1}} + \sum_{h=1}^H \left(\text{Head}_h^l(z_{l-1}) - \overline{\text{Head}_h^l(\overline{z_{l-1}})} \right) \right\|_F \\ \|Y_l - \overline{Y_l}\|_F &\leq \|E_{l-1}\|_F + \sum_{h=1}^H (\beta \mathcal{D}_{att,l}^h + \Psi_l^h \|E_{l-1}\|_F) \\ \|Y_l - \overline{Y_l}\|_F &\leq \beta \sum_{h=1}^H \mathcal{D}_{att,l}^h + \left(1 + \sum_{h=1}^H \Psi_l^h \right) \|E_{l-1}\|_F \end{aligned}$$

As mentioned above, the output error of the FFN is bounded by

$$\|E_l\|_F \leq \beta (\mathcal{T}_{2,l} + \left\| \overline{W_{2,l}} \right\|_2 \mathcal{T}_{1,l}) + \left\| \overline{W_{2,l}} \right\|_2 \cdot \left\| \overline{W_{1,l}} \right\|_2 \cdot \|Y_l - \overline{Y_l}\|_F$$

Substitute the bound for $\|Y_l - \overline{Y_l}\|_F$ into the equation:

$$\|E_l\|_F \leq \beta \mathcal{T}_{2,l} + \beta \left\| \overline{W_{2,l}} \right\|_2 \mathcal{T}_{1,l} + \left\| \overline{W_{2,l}} \right\|_2 \left\| \overline{W_{1,l}} \right\|_2 \left[\beta \sum_{h=1}^H \mathcal{D}_{att,l}^h + \left(1 + \sum_{h=1}^H \Psi_l^h \right) \|E_{l-1}\|_F \right]$$

Group the terms into the local layer distortion $\mathcal{D}_l$ and the layer Lipschitz amplification Λ_l :

$$\begin{aligned} \|E_l\|_F &\leq \underbrace{\beta \left[\mathcal{T}_{2,l} + \left\| \overline{W_{2,l}} \right\|_2 \mathcal{T}_{1,l} + \left\| \overline{W_{2,l}} \right\|_2 \left\| \overline{W_{1,l}} \right\|_2 \sum_{h=1}^H \mathcal{D}_{att,l}^h \right]}_{\mathcal{D}_l} + \underbrace{\left[\left\| \overline{W_{2,l}} \right\|_2 \left\| \overline{W_{1,l}} \right\|_2 \left(1 + \sum_{h=1}^H \Psi_l^h \right) \right]}_{\Lambda_l} \|E_{l-1}\|_F \\ \|E_l\|_F &\leq \beta \mathcal{D}_l + \Lambda_l \|E_{l-1}\|_F \end{aligned}$$

By unrolling the recurrence, considering $\|E_0\|_F = 0$:

$$\begin{aligned} \mathbb{E}\|E_L\|_F &\leq \mathbb{E} \left[\sum_{l=1}^L \left(\prod_{k=l+1}^L \Lambda_k \right) \beta \mathcal{D}_l \right] \\ \mathbb{E}\|f(X) - \tilde{f}(X)\|_F &\leq \beta \sum_{l=1}^L \left(\prod_{k=l+1}^L \Lambda_k \right) \mathcal{D}_l \end{aligned}$$

□

While the explicit algebraic expansion relies on sequential worst-case multiplicative error accumulations, rendering the numerical bound loose in practical high-dimensional settings, it establishes a rigorous theoretical linkage between stable rank and deep network capacity. We have successfully traced the SR-constrained matrix error $\mathcal{T}$ from a single linear projection through the highly non-linear, dual-path architecture of a multi-head transformer.

The theorem formalizes how applying LoRA to the Query, Key, Value, and Output projection matrices restricts the adaptability of the attention head according to the stable rank limit. Even though the frozen base matrices retain their full rank, steering the attention mechanism toward a significantly different target function might require a high-SR matrix. If the downstream data intrinsically demands high stable rank updates to form new associations between tokens, an optimization trajectory that implicitly biases toward low sr solutions will fundamentally cap the Transformer’s representational accuracy. This mathematical bottleneck suggests a strong hypothesis for the performance plateaus often observed empirically at high nominal ranks: real expressivity scales primarily with stable rank rather than algebraic rank, meaning that simply increasing the parameter budget r yields diminishing returns if the effective dimension of the update remains constricted by the optimization algorithm.

3.5 Generalization Analysis

Understanding generalization – why a model performs well on unseen data despite being most of the times massively overparameterized – is a central pursuit in deep learning theory. The work of Kratsios et al. [15] recently established generalization bounds for low-rank adapters in terms of rank and sample complexity. Their upper bound scales as $\mathcal{O}(\sqrt{r/N})$, where r is the nominal adapter rank and N is the sample size. This bound, derived from worst-case estimates of rank- r matrices, theoretically suggests that generalization degrades proportionally as one increases the nominal rank.

However, this analysis relies purely on structural constraints and does not account for the continuous spectral structure or energy distribution of the learned adapters. The gap between Kratsios et al.’s bound and empirical performance (where larger ranks often do not overfit as aggressively as predicted) stems directly from their worst-case analysis assumptions. Bounds relying purely on algebraic rank are mathematically tight only when all r singular values are roughly equal ($\sigma_1 = \sigma_2 = \dots = \sigma_r$), representing the maximum entropy distribution under a rank constraint. Because LoRA optimization empirically collapses into highly skewed distributions with low stable rank, evaluating complexity purely by nominal rank vastly overestimates the capacity of the model.

In this section, we rigorously prove that the generalization bound can be significantly tightened by directly substituting nominal rank with stable rank, leveraging the PAC-Bayesian framework.

3.5.1 PAC-Bayesian Framework Integration

The PAC-Bayesian framework [24, 23, 33] provides an elegant and powerful tool for deriving generalization bounds. Unlike VC-dimension or Rademacher complexity bounds—which count combinatorial capacity—PAC-Bayes provides data-dependent bounds that can natively incorporate prior knowledge about the hypothesis space. This is uniquely crucial for analyzing fine-tuning, where the massive pretrained model acts as a highly informative prior.

The key mechanism bounds the generalization gap through the Kullback-Leibler (KL) divergence $KL(\mathcal{Q}||\mathcal{P})$ between a posterior distribution $\mathcal{Q}$ (distribution over fine-tuned weights) and a prior $\mathcal{P}$ (distribution over pretrained weights). For LoRA adaptation, if we set $\mathcal{P} = \prod_t \mathcal{N}(W_{0,t}, \tau_t^2 I)$ and $\mathcal{Q} = \prod_t \mathcal{N}(W_{0,t} + \Delta W_t, \tau_t^2 I)$, the KL divergence cleanly reduces to a sum of squared Frobenius norms: $KL(\mathcal{Q}||\mathcal{P}) = \sum_t \frac{\|\Delta W_t\|_F^2}{2\tau_t^2}$.

While previous works [2, 27] have shown that generalization depends on the product of spectral norms across layers, our key insight is that the Frobenius norm inherently decomposes as $\|\Delta W\|_F^2 = \|\Delta W\|_{op}^2 \cdot \text{sr}(\Delta W)$. This exact identity allows us to substitute the worst-case bound $\|\Delta W\|_F^2 \leq \Lambda^2 r$ with the precise bound $\|\Delta W\|_F^2 \leq \Lambda^2 S$. This provides a direct, continuous link between the stable rank of the adapter and the network’s generalization capability.

3.5.2 Notation

In this section we use a setting similar to the one described in [15]. Consider a depth- T linear network f_{W_0} with 1-Lipschitz activation functions $\sigma(\cdot)$ such that $\sigma(0) = 0$, pre-trained weights W_0 and $W_{0,t} \in \mathbb{R}^{d_{t-1} \times d_t}$ weight matrices for each layer. LoRA is applied via matrices $\Delta W_t = B_t(A_t)^T$, so that $W_t = W_{0,t} + \Delta W_t$. We denote the resulting network f_W .

We consider the multi-label classification problem with unknown data distribution $\mathcal{D}$ and the set of i.i.d. samples $S = \{(x_1, y_1), \dots, (x_N, y_N)\}$, such that $\|x_i\|_2 \leq B_x$, $y_i \in \{1, \dots, K\}$. For input x we also denote intermediate states $h_0 = x$, $h_t = \sigma(W_t h_{t-1})$, $f_W(x) = h_T$.

Additionally, we require limited spectral norm for the network before and after the adaptation: $\|W_{0,t}\|_{op} \leq B_t$, $\|\Delta W_t\|_{op} \leq \Lambda_t$, $\|W\|_{op} \leq \bar{\beta}_t = B_t + \Lambda_t$ and assume, that stable rank of adapters is bounded by some constant $\text{sr}(\Delta W_t) \leq S_t$. Maximum layer width: $d_{max} = \max_t(d_t)$. The maximum number of parameters in layer: $D_{max} = \max_t(d_t \cdot d_{t-1})$.

For the main result in this section we use the Margin Loss. First we define the margin m for model f_W and data point (x, y) as follows:

$$m(f_W(x), y) = f_W(x)_y - \max_{j \neq y} f_W(x)_j$$

Then the true γ -margin error for f_W :

$$L_\gamma(W) = \mathbb{P}_{(x,y)} [m(f_W(x), y) \leq \gamma]$$

Then L_0 is the expected incorrect classification probability for the network or the true risk function. And the empirical γ -margin error:

$$\hat{L}_\gamma(W) = \frac{1}{N} \sum_{i=1}^N \mathbb{I} [m(f_W(x_i), y_i) \leq \gamma]$$

3.5.3 Upper-bound for LoRA Generalization Gap

The following theorem formally derives the generalization bound incorporating the layer-wise stable rank constants S_t . The result isolates exactly how reducing stable rank acts as a powerful regularizer, naturally minimizing the generalization gap independently of the nominal dimensions.

Theorem 3.

For any $\gamma > 0$ and any $W = W_0 + \Delta W$ that satisfies the aforementioned conditions with probability at least $1 - \delta$ by the choice of training sample S . The true classification risk is bounded by the following:

$$L_0(W) \leq \hat{L}_\gamma(W) + \mathcal{O} \left(\sqrt{\frac{B_x^2 T^2 (d_{max} + \ln(TN)) \left(\prod_{t=1}^T \bar{\beta}_t^2 \right) \sum_{l=1}^L \frac{\Lambda_l^2 S_l}{\bar{\beta}_l^2} + \log \left(\frac{N}{\delta} \right)}{\gamma^2 N}} \right)$$

Доказательство.

We start with the fundamental PAC-Bayes theorem [24][23]:

For any prior predictor distribution $\mathcal{P}$ (not depending on S) and any $\delta \in (0, 1)$, with probability at least $1 - \delta$, for any predictor distribution $\mathcal{Q}$ (depending on S) and the margin $m = \gamma/2$ error function the following holds:

$$\mathbb{E}_{w \sim \mathcal{Q}} [L_{\gamma/2}(w)] \leq \mathbb{E}_{w \sim \mathcal{Q}} [\hat{L}_{\gamma/2}(w)] + \sqrt{\frac{KL(\mathcal{Q}||\mathcal{P}) + \log \left(\frac{2N}{\delta} \right)}{2N}}$$

Now we need to choose the distributions to be able to remove the expectations $\mathbb{E}_{w \sim \mathcal{Q}}$.

Let $\mathcal{Q}$ be a distribution such that layer t weights matrix is $w_t = W_t + U_t$ and $U_t \sim \mathcal{N}(0, \tau_t^2 I)$. The exact value of variance τ_t^2 is to be determined.

For given noise sample U consider the maximum output perturbation over the dataset:

$$\Delta f(U) = \max_{x \in S} \|f_{W+U}(x) - f_W(x)\|_\infty$$

Assume $\Delta f(U) \leq \gamma/4$, then for any (x, y) :

$$|m(f_W(x), y) - m(f_{W+U}(x), y)| \leq \gamma/2$$

Thus, if f_W makes mistake $m(f_W(x), y) \leq 0$, then $m(f_{W+U}(x), y) \leq \gamma/2$ which results in the following risks inequality:

$$L_0(W) \leq L_{\gamma/2}(W + U)$$

Similarly, the following holds:

$$\hat{L}_{\gamma/2}(W + U) \leq \hat{L}_\gamma(W)$$

So, combining with the assumption we get

$$\mathbb{I}[m(f_W(x), y) \leq 0] \leq \mathbb{I}[m(f_{W+U}(x), y) \leq \gamma/2] + \mathbb{I}[\Delta f(U) > \gamma/4]$$

and after taking the expectation:

$$L_0(W) \leq \mathbb{E}_U[L_{\gamma/2}(W + U)] + \mathbb{P}_U[\Delta f(U) > \gamma/4]$$

by doing the same, but averaging over S instead of taking the expectation we get:

$$\mathbb{I}[m(f_{W+U}(x), y) \leq \gamma/2] \leq \mathbb{I}[m(f_W(x), y) \leq \gamma] + \mathbb{I}[\Delta f(U) > \gamma/4]$$

$$\mathbb{E}_U[\hat{L}_{\gamma/2}(W + U)] \leq \hat{L}_\gamma(W) + \mathbb{P}_U[\Delta f(U) > \gamma/4]$$

Here we substitute these inequalities into the fundamental theorem:

$$L_0(W) \leq \mathbb{E}_U[L_{\gamma/2}(W + U)] + \mathbb{P}_U[\Delta f(U) > \gamma/4] \leq \mathbb{E}_U[\hat{L}_{\gamma/2}(W + U)] + \mathbb{P}_U[\Delta f(U) > \gamma/4] + \sqrt{\frac{KL(\mathcal{Q}||\mathcal{P}) + \log(\frac{2N}{\delta})}{2N}}$$

$$L_0(W) \leq \hat{L}_\gamma(W) + 2\mathbb{P}_U[\Delta f(U) > \gamma/4] + \sqrt{\frac{KL(\mathcal{Q}||\mathcal{P}) + \log(\frac{2N}{\delta})}{2N}}$$

Now we must establish a strict upper bound on the perturbation $\Delta f(U)$ to determine the appropriate variance τ_t^2 that makes $\mathbb{P}_U[\Delta f(U) > \gamma/4]$ negligibly small.

Bounding the network perturbation:

Let $h_t(x)$ denote the activations of the deterministic network f_W at layer t , and $\tilde{h}_t(x)$ denote the activations of the noisy network f_{W+U} , let $h_0(x) = \tilde{h}_0(x) = x$.

For any subsequent layer t :

$$\|\tilde{h}_t - h_t\|_2 = \|\sigma((W_t + U_t)\tilde{h}_{t-1}) - \sigma(W_t h_{t-1})\|_2$$

Using the 1-Lipschitz property of the activation function σ and the triangle inequality, we get:

$$\|\tilde{h}_t - h_t\|_2 \leq \|(W_t + U_t)\tilde{h}_{t-1} - W_t h_{t-1}\|_2 \leq \|W_t\|_{op} \|\tilde{h}_{t-1} - h_{t-1}\|_2 + \|U_t\|_{op} \|\tilde{h}_{t-1}\|_2$$

Since $\|W_t\|_{op} \leq \bar{\beta}_t$, we obtain a recursive inequality:

$$\|\tilde{h}_t - h_t\|_2 \leq \bar{\beta}_t \|\tilde{h}_{t-1} - h_{t-1}\|_2 + \|U_t\|_{op} \|\tilde{h}_{t-1}\|_2$$

Unrolling this recursion from layer T down to layer 1, we obtain the bound for the final network output:

$$\|f_{W+U}(x) - f_W(x)\|_2 = \|\tilde{h}_T - h_T\|_2 \leq \sum_{t=1}^T \left(\prod_{i=t+1}^T \bar{\beta}_i \right) \|U_t\|_{op} \|\tilde{h}_{t-1}\|_2$$

To bound the norm of the noisy activations $\|\tilde{h}_{t-1}\|_2$, we do the following:

$$\|\tilde{h}_{t-1}\|_2 \leq \|x\|_2 \prod_{i=1}^{t-1} \|W_i + U_i\|_{op} \leq B_x \prod_{i=1}^{t-1} (\bar{\beta}_i + \|U_i\|_{op})$$

Assuming that we can choose the noise variance small enough (we do that below) such that for all layers $\|U_i\|_{op} \leq \frac{1}{T} \bar{\beta}_i$, we can bound the product:

$$\begin{aligned} \bar{\beta}_i + \|U_i\|_{op} &\leq \bar{\beta}_i \left(1 + \frac{1}{T}\right) \\ \|\tilde{h}_{t-1}\|_2 &\leq B_x \left(1 + \frac{1}{T}\right)^{t-1} \prod_{i=1}^{t-1} \bar{\beta}_i \leq e B_x \prod_{i=1}^{t-1} \bar{\beta}_i \end{aligned}$$

Substituting this back into the unrolled recursion, we get:

$$\|f_{W+U}(x) - f_W(x)\|_2 \leq \sum_{t=1}^T \left(\prod_{i=t+1}^T \bar{\beta}_i \right) \|U_t\|_{op} \left(e B_x \prod_{i=1}^{t-1} \bar{\beta}_i \right) = e B_x \left(\prod_{i=1}^T \bar{\beta}_i \right) \sum_{t=1}^T \frac{\|U_t\|_{op}}{\bar{\beta}_t}$$

Since we bounded $\|x\|_2$ with B_x we have already taken the maximum over $x \in S$, so the final perturbation bound $\Delta f(U)$:

$$\Delta f(U) \leq e B_x \left(\prod_{i=1}^T \bar{\beta}_i \right) \sum_{t=1}^T \frac{\|U_t\|_{op}}{\bar{\beta}_t}$$

Bounding the spectral norm of U :

Since the noise matrix $U_t \in \mathbb{R}^{d_t \times d_{t-1}}$ has entries from $\mathcal{N}(0, \tau_t^2)$ we can apply Gordon's theorem for gaussian matrices[40], for any $s > 0$, with probability at least $1 - \exp(-s^2/2)$:

$$\|U_t\|_{op} \leq \tau_t (\sqrt{d_t} + \sqrt{d_{t-1}} + s) \leq \tau_t (2\sqrt{d_{max}} + s)$$

For all T layers simultanously this hold with probability at least $1 - T \exp(-s^2/2)$.

Chosing the variance and bounding the failure probability:

As mentioned above, we require the total perturbation to satisfy $\Delta f(U) \leq \gamma/4$. Substituting the spectral norm bound into the perturbation inequality yields:

$$\Delta f(U) \leq e B_x \left(\prod_{i=1}^T \bar{\beta}_i \right) \sum_{t=1}^T \frac{\tau_t (2\sqrt{d_{max}} + s)}{\bar{\beta}_t}$$

To balance the noise across layers, we set the variance multiplier proportional to the spectral norm bound of each layer: $\tau_t = \tau \bar{\beta}_t$. The sum then simplifies:

$$\sum_{t=1}^T \frac{\tau \bar{\beta}_t (2\sqrt{d_{max}} + s)}{\bar{\beta}_t} = \tau T (2\sqrt{d_{max}} + s)$$

Thus, the perturbation is bounded by:

$$\Delta f(U) \leq e B_x \left(\prod_{i=1}^T \bar{\beta}_i \right) \tau T (2\sqrt{d_{max}} + s)$$

We set this expression exactly equal to $\gamma/4$ and solve for the global variance data dependent parameter τ^* :

$$\tau^* = \frac{\gamma}{4e B_x T (2\sqrt{d_{max}} + s) \prod_{i=1}^T \bar{\beta}_i}$$

Now, we additionally require the probability of the perturbation exceeding $\gamma/4$ to be negligibly small, specifically

$$\mathbb{P}_U[\Delta f(U) > \gamma/4] \leq \frac{1}{\sqrt{N}}$$

Setting the failure probability from the $\|U\|_{op}$ bound equal to this value to find the specific s :

$$T \exp(-s^2/2) = \frac{1}{\sqrt{N}} \implies s = \sqrt{2 \ln(T\sqrt{N})}$$

Calculating KL Divergence and Introducing Stable Rank:

Let the distribution $\mathcal{P}$ be the $\prod_{t=1}^T \mathcal{N}(W_{0,t}, \tau_t^2 I)$. Distribution $\mathcal{Q}$ as previously mention is $\prod_{t=1}^T \mathcal{N}(W_{0,t} + \Delta W_t, \tau_t^2 I)$. The KL divergence between these isotropic multivariate gaussians is the following:

$$KL(\mathcal{Q}||\mathcal{P}) = \sum_{t=1}^T \frac{\|(W_{0,t} + \Delta W_t) - W_{0,t}\|_F^2}{2\tau_t^2} = \sum_{t=1}^T \frac{\|\Delta W_t\|_F^2}{2\tau_t^2}$$

Substituting $\tau_t^2 = \tau^2 \bar{\beta}_t^2$, we obtain:

$$KL(\mathcal{Q}||\mathcal{P}) = \frac{1}{2\tau^2} \sum_{t=1}^T \frac{\|\Delta W_t\|_F^2}{\bar{\beta}_t^2}$$

Here we finally utilize the definition of stable rank: $\text{sr}(\Delta W_t) = \frac{\|\Delta W_t\|_F^2}{\|\Delta W_t\|_{op}^2} \leq S_t$. Since the spectral norm of the adapter is bounded by $\|\Delta W_t\|_{op} \leq \Lambda_t$, then the Frobenius norm is bounded by:

$$\|\Delta W_t\|_F^2 \leq \Lambda_t^2 S_t$$

Substituting this inequality into the KL divergence yields:

$$KL(\mathcal{Q}||\mathcal{P}) \leq \frac{1}{2\tau^2} \sum_{t=1}^T \frac{\Lambda_t^2 S_t}{\bar{\beta}_t^2}$$

The union bound over τ (PAC-Bayes prior fixing):

A fundamental requirement of the PAC-Bayes framework is that the prior distribution $\mathcal{P}$ (and consequently τ) must be chosen before observing the training data S . However, the optimal variance τ^* depends on the spectral norms $\bar{\beta}_t$ and the actual learned parameters, making it data-dependent. We cannot directly use τ^* to define a single prior without violating the PAC-Bayes independence assumption.

To resolve this rigorously, we employ a standard union bound technique over a predefined grid of priors. Before observing the data, we define a countably infinite grid of variance multipliers:

$$\tau_{(k)} = \exp(k/T), \quad \text{for } k \in \mathbb{Z}$$

For each k , we construct a specific prior $\mathcal{P}_k$ characterized by the layer-wise variances $\tau_{t,(k)}^2 = \tau_{(k)}^2 \bar{\beta}_t^2$.

We must distribute our total allowed failure probability δ across all these priors. We assign a confidence level δ_k to the k -th prior such that:

$$\delta_k = \frac{\delta}{2(|k|+1)^2}, \quad \sum_{k=-\infty}^{\infty} \delta_k \leq \delta$$

By applying the union bound, the fundamental PAC-Bayes inequality (established at the beginning of the proof) holds simultaneously for all priors $\mathcal{P}_k \in \{\mathcal{P}_k\}_{k \in \mathbb{Z}}$ with probability at least $1 - \delta$.

After the training process is complete and the adapters ΔW_t are obtained, we calculate the theoretically optimal data-dependent variance τ^* . We then simply select the integer index k such that our grid value closely approximates the optimal value:

$$\tau_{(k)} \leq \tau^* < \tau_{(k+1)}$$

Because $\tau_{(k+1)}/\tau_{(k)} = \exp(1/T)$, it is guaranteed that $\tau_{(k)} > \tau^* e^{-1/T}$.

Substituting this into KL -divergence considering $T \geq 1 \implies e^{2/T} \leq e^2$, yields the following bound:

$$KL(\mathcal{Q}||\mathcal{P}_k) = \frac{1}{2\tau_{(k)}^2} \sum_{t=1}^T \frac{\|\Delta W_t\|_F^2}{\bar{\beta}_t^2} \leq \frac{e^2}{2(\tau^*)^2} \sum_{t=1}^T \frac{\Lambda_t^2 S_t}{\bar{\beta}_t^2}$$

Furthermore, in the PAC-Bayes bound, the logarithmic term now uses δ_k instead of δ :

$$\log\left(\frac{2N}{\delta_k}\right) = \log\left(\frac{2N}{\delta}\right) + \log(2(|k| + 1)^2)$$

To understand the magnitude of this penalty, recall that we chose k such that $\exp(k/T) \approx \tau^*$. Therefore, the index magnitude $|k|$ is bounded by $\mathcal{O}(T|\ln(\tau^*)|)$. Looking at the closed-form expression for τ^* , $|\ln(\tau^*)|$ scales logarithmically with the network dimensions and the product of spectral norms $|\ln(\tau^*)| = \mathcal{O}(\ln(d_{max}) + \sum_{i=1}^T \ln(\bar{\beta}_i))$.

Consequently, the penalty term $\log(2(|k| + 1)^2)$ introduces a double-logarithmic dependence on the network parameters and norms:

$$\log(2(|k| + 1)^2) = \mathcal{O}\left(\log\left(T \ln(d_{max}) + T \sum_{i=1}^T \ln(\bar{\beta}_i)\right)\right)$$

This $\log \log(\cdot)$ penalty grows so slowly that it behaves as a negligible $\tilde{\mathcal{O}}(1)$ constant for any practically realizable neural network.

Final substitution and asymptotic bound:

Now we substitute the expression for τ^* into our bounded KL divergence:

$$\frac{1}{(\tau^*)^2} = \frac{16e^2 B_x^2 T^2 \left(2\sqrt{d_{max}} + \sqrt{2\ln(T\sqrt{N})}\right)^2 \prod_{t=1}^T \bar{\beta}_t^2}{\gamma^2}$$

Using the basic algebraic inequality $(a + b)^2 \leq 2a^2 + 2b^2$, we obtain the following:

$$\left(2\sqrt{d_{max}} + \sqrt{2\ln(T\sqrt{N})}\right)^2 \leq 8d_{max} + 4\ln(T\sqrt{N})$$

As $N \geq 1$, we can derive another bound:

$$\ln(T\sqrt{N}) = \ln(T) + \frac{1}{2} \ln(N) \leq \ln(TN)$$

Thus, the KL divergence is then asymptotically bounded by:

$$KL(\mathcal{Q}||\mathcal{P}_k) = \mathcal{O}\left(\frac{B_x^2 T^2 (d_{max} + \ln(TN)) \left(\prod_{t=1}^T \bar{\beta}_t^2\right)}{\gamma^2} \sum_{t=1}^T \frac{\Lambda_t^2 S_t}{\bar{\beta}_t^2}\right)$$

Finally, we substitute the KL bound and our previously fixed $\mathbb{P}_U \leq 1/\sqrt{N}$ back into the fundamental PAC-Bayes inequality established at the beginning of the proof. Since the term $2/\sqrt{N} = \mathcal{O}(1/\sqrt{N})$ scales identically to the denominator of the square root, it is cleanly absorbed into the main $\mathcal{O}(\cdot)$ notation. Taking the square root of the KL divergence expression yields the final bound:

$$L_0(W) \leq \hat{L}_\gamma(W) + \mathcal{O}\left(\sqrt{\frac{B_x^2 T^2 (d_{max} + \ln(TN)) \left(\prod_{t=1}^T \bar{\beta}_t^2\right) \sum_{t=1}^T \frac{\Lambda_t^2 S_t}{\bar{\beta}_t^2} + \log\left(\frac{N}{\delta}\right)}{\gamma^2 N}}\right)$$

This completes the proof. □

Theorem 3 demonstrates mathematically why tuning large LoRA ranks rarely leads to catastrophic overfitting, despite adding millions of parameters. The generalization gap correlates tightly with the stable rank sum $\sum_t \Lambda_t^2 S_t / \bar{\beta}_t^2$. Even if the nominal rank r is arbitrarily large, if the training dynamics implicitly bias the updates toward highly skewed singular value distributions (meaning $S_t \ll r$), the PAC-Bayesian bound naturally absorbs this regularization. Furthermore, the term $\Lambda_t^2 / \bar{\beta}_t^2$ measures the relative magnitude of the adapter perturbation compared to the total layer norm. This indicates that as long as the adapter is a relatively small additive correction with highly concentrated energy, the generalization gap remains constrained.

4 Empirical Validation

4.1 Toy experiment details

Рис. 1: Experiment A: LoRA’s implicit bias toward low stable rank. Left: Training loss decreases with rank. Right: Stable rank evolution shows all LoRA models converge below target SR 38.17, with $r = 256$ reaching only SR 31.26 while Full FT reaches 38.17.

Рис. 2: Experiment C: Stable rank as universal generalization predictor. Panel 3 shows linear relationship $\Delta_{gen} \propto \sqrt{\text{sr}(\Delta W)}$ across all conditions, with Full FT points aligning on the same line. Panel 4 shows nominal rank alone fails to predict generalization, as SR plateaus with increasing rank.

4.2 LLM Experiments

4.3 Analysis

5 Discussion

6 Conclusion

In this thesis, we established stable rank as the mathematically appropriate theoretical measure for analyzing LoRA’s properties, correcting the persistent reliance on algebraic rank in existing literature. Our key contributions are:

Рис. 3: Per-layer stable rank for Qwen2.5-0.5B with LoRA (rank 64). Stable rank is consistently 2-5 across all layers, significantly lower than nominal rank 64, validating that our theoretical bounds using SR are practically relevant.

Fundamental Matrix Limits (Lemmas 1-2): We derived strict bounds on matrix approximation under stable rank constraints, demonstrating an irreducible Frobenius error when attempting to approximate high-SR targets with low-SR matrices. We provided a constructive optimal bound and the way to reach it.

Expressivity Bounds (Theorems 1-2): We extended these single-matrix limits to Deep Neural Networks. For FNNs, we proved how the layer-wise SR error $\mathcal{T}_l$ amplifies multiplicatively through network depth. For Transformers, we decomposed the constraint into FFN parameter errors, attention head distortions, and softmax score sensitivities, proving that low-SR attention projections fundamentally cap expressivity.

Generalization Bounds (Theorem 3): We leveraged a PAC-Bayesian framework to rigorously prove a tighter generalization gap that scales with $\mathcal{O}(\sqrt{\sum_t \Lambda_t^2 S_t / \beta_t^2})$. Because optimization implicitly biases LoRA such that $S_t \ll r_t$, our theorem mathematically validates why massively overparameterized LoRA adapters generalize remarkably well in practice.

Список литературы

- [1] Anonymous Authors. Bernoulli-lora: A theoretical framework for randomized low-rank adaptation. arXiv preprint, 2025.
- [2] Peter L Bartlett, Dylan J Foster, and Matus J Telgarsky. Spectrally-normalized margin bounds for neural networks. *Advances in Neural Information Processing Systems*, 30, 2017.
- [3] Dan Biderman, Jose Javier Gonzalez Ortiz, Jacob Portes, Mansheej Paul, Philip Greengard, Connor Jennings, et al. Lora learns less and forgets less. *Transactions on Machine Learning Research*, 2024.
- [4] DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, Xiaokang Zhang, Xingkai Yu, Yu Wu, Z. F. Wu, Zhibin Gou, Zhihong Shao, Zhuoshu Li, Ziyi Gao, Aixin Liu, Bing Xue, Bingxuan Wang, Bochao Wu, Bei Feng, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, Damai Dai, Deli Chen, Dongjie Ji, Erhang Li, Fangyun Lin, Fucong Dai, Fuli Luo, Guangbo Hao, Guanting Chen, Guowei Li, H. Zhang, Han Bao, Hanwei Xu, Haocheng Wang, Honghui Ding, Huajian Xin, Huazuo Gao, Hui Qu, Hui Li, Jianzhong Guo, Jiashi Li, Jiawei Wang, Jingchang Chen, Jingyang Yuan, Junjie Qiu, Junlong Li, J. L. Cai, Jiaqi Ni, Jian Liang, Jin Chen, Kai Dong, Kai Hu, Kaige Gao, Kang Guan, Kexin Huang, Kuai Yu, Lean Wang, Lecong Zhang, Liang Zhao, Litong Wang, Liyue Zhang, Lei Xu, Leyi Xia, Mingchuan Zhang, Minghua Zhang, Minghui Tang, Meng Li, Miaojuan Wang, Mingming Li, Ning Tian, Panpan Huang, Peng Zhang, Qiancheng Wang, Qinyu Chen, Qiusi Du, Ruiqi Ge, Ruisong Zhang, Ruizhe Pan, Runji Wang, R. J. Chen, R. L. Jin, Ruyi Chen, Shanghao Lu, Shangyan Zhou, Shanhuang Chen, Shengfeng Ye,

- Shiyu Wang, Shuiping Yu, Shunfeng Zhou, Shuting Pan, S. S. Li, Shuang Zhou, Shaoqing Wu, Shengfeng Ye, Tao Yun, Tian Pei, Tianyu Sun, T. Wang, Wangding Zeng, Wanjia Zhao, Wen Liu, Wenfeng Liang, Wenjun Gao, Wenqin Yu, Wentao Zhang, W. L. Xiao, Wei An, Xiaodong Liu, Xiaohan Wang, Xiaokang Chen, Xiaotao Nie, Xin Cheng, Xin Liu, Xin Xie, Xingchao Liu, Xinyu Yang, Xinyuan Li, Xuecheng Su, Xuheng Lin, X. Q. Li, Xiangyue Jin, Xiaojin Shen, Xiaosha Chen, Xiaowen Sun, Xiaoxiang Wang, Xinnan Song, Xinyi Zhou, Xianzu Wang, Xinxia Shan, Y. K. Li, Y. Q. Wang, Y. X. Wei, Yang Zhang, Yanhong Xu, Yao Li, Yao Zhao, Yaofeng Sun, Yaohui Wang, Yi Yu, Yichao Zhang, Yifan Shi, Yiliang Xiong, Ying He, Yishi Piao, Yisong Wang, Yixuan Tan, Yiyang Ma, Yiyuan Liu, Yongqiang Guo, Yuan Ou, Yuduan Wang, Yue Gong, Yuheng Zou, Yujia He, Yunfan Xiong, Yuxiang Luo, Yuxiang You, Yuxuan Liu, Yuyang Zhou, Y. X. Zhu, Yanhong Xu, Yanping Huang, Yaohui Li, Yi Zheng, Yuchen Zhu, Yunxian Ma, Ying Tang, Yukun Zha, Yuting Yan, Z. Z. Ren, Zehui Ren, Zhangli Sha, Zhe Fu, Zhean Xu, Zhenda Xie, Zhengyan Zhang, Zhewen Hao, Zhicheng Ma, Zhigang Yan, Zhiyu Wu, Zihui Gu, Zijia Zhu, Zijun Liu, Zilin Li, Ziwei Xie, Ziyang Song, Zizheng Pan, Zhen Huang, Zhipeng Xu, Zhongyu Zhang, and Zhen Zhang. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning, 2025.
- [5] Ning Ding, Yujia Qin, Guang Yang, Fuchao Wei, Zonghan Yang, Yusheng Su, Shengding Hu, Yulin Chen, Chi-Min Chan, Weize Chen, et al. Parameter-efficient fine-tuning of large-scale pre-trained language models. *Nature Machine Intelligence*, 5(3):220–235, 2023.
- [6] Carl Eckart and Gale Young. The approximation of one matrix by another of lower rank. *Psychometrika*, 1(3):211–218, 1936.
- [7] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, Aurelien Rodriguez, Austen Gregerson, Ava Spataru, Baptiste Roziere, Bethany Biron, Binh Tang, Bobbie Chern, Charlotte Caucheteux, Chaya Nayak, Chloe Bi, Chris Marra, Chris McConnell, Christian Keller, Christophe Touret, Chunyang Wu, Corinne Wong, Cristian Canton Ferrer, Cyrus Nikolaidis, Damien Allonsius, Daniel Song, Danielle Pintz, Danny Livshits, Danny Wyatt, David Esiobu, Dhruv Choudhary, Dhruv Mahajan, Diego Garcia-Olano, Diego Perino, Dieuwke Hupkes, Egor Lakomkin, Ehab AlBadawy, Elina Lobanova, Emily Dinan, Eric Michael Smith, Filip Radenovic, Francisco Guzmán, Frank Zhang, Gabriel Synnaeve, Gabrielle Lee, Georgia Lewis Anderson, Govind Thattai, Graeme Nail, Gregoire Mialon, Guan Pang, Guillem Cucurell, Hailey Nguyen, Hannah Korevaar, Hu Xu, Hugo Touvron, Iliyan Zarov, Imanol Arrieta Ibarra, Isabel Kloumann, Ishan Misra, Ivan Evtimov, Jack Zhang, Jade Copet, Jaewon Lee, Jan Geffert, Jana Vranes, Jason Park, Jay Mahadeokar, Jeet Shah, Jelmer van der Linde, Jennifer Billock, Jenny Hong, Jenya Lee, Jeremy Fu, Jianfeng Chi, Jianyu Huang, Jiawen Liu, Jie Wang, Jiecao Yu, Joanna Bitton, Joe Spisak, Jongsoo Park, Joseph Rocca, Joshua Johnstun, Joshua Saxe, Junteng Jia, Kalyan Vasuden Alwala, Karthik Prasad, Kartikeya Upasani, Kate Plawiak, Ke Li, Kenneth Heafield, Kevin Stone, Khalid El-Arini, Krithika Iyer, Kshitiz Malik, Kuenley Chiu, Kunal Bhalla, Kushal Lakhotia, Lauren Rantala-Yeary, Laurens van der Maaten, Lawrence Chen, Liang Tan, Liz Jenkins, Louis Martin, Lovish Madaan, Lubo Malo, Lukas Blecher, Lukas Landzaat, Luke de Oliveira, Madeline Muzzi, Mahesh Pasupuleti, Mannat Singh, Manohar Paluri, Marcin Kardas, Maria Tsimpoukelli, Mathew Oldham, Mathieu Rita, Maya Pavlova, Melanie Kambadur, Mike Lewis, Min Si, Mitesh Kumar Singh, Mona Hassan, Naman Goyal, Narjes Torabi, Nikolay Bashlykov, Nikolay Bogoychev, Niladri Chatterji, Ning Zhang, Olivier Duchenne, Onur Çelebi, Patrick Alrassy, Pengchuan Zhang, Pengwei Li, Petar Vasic, Peter Weng, Prajjwal Bhargava, Pratik Dubal, Praveen Krishnan, Punit Singh Koura, Puxin Xu, Qing He, Qingxiao Dong, Ragavan Srinivasan, Raj Ganapathy, Ramon Calderer, Ricardo Silveira Cabral, Robert Stojnic, Roberta Raileanu, Rohan Maheswari, Rohit Girdhar, Rohit Patel, Romain Sauvestre, Ronnie Polidoro, Roshan Sumbaly, Ross Taylor, Ruan Silva, Rui Hou, Rui Wang, Saghar Hosseini, Sahana Chennabasappa, Sanjay Singh, Sean Bell, Seohyun Sonia Kim, Sergey Edunov, Shaoqiang Nie, Sharan Narang, Sharath Raparthy, Sheng Shen, Shengye Wan, Shruti Bhosale, Shun Zhang, Simon Vandenhende, Soumya Batra, Spencer Whitman, Sten Sootla, Stephane Collot, Suchin Gururangan, Sydney Borodinsky, Tamar Herman, Tara Fowler, Tarek Sheasha, Thomas Georgiou, Thomas Scialom, Tobias Speckbacher, Todor Mihaylov, Tong Xiao, Ujjwal Karn, Vedanuj Goswami, Vibhor Gupta, Vignesh Ramanathan, Viktor Kerkez, Vincent Gonguet, Virginie Do, Vish Vogeti, Vitor Albiero, Vladan Petrovic, Weiwei Chu, Wenhan Xiong, Wenyan Fu, Whitney Meers, Xavier Martinet, Xiaodong Wang, Xiaofang Wang, Xiaoqing Ellen Tan, Xide Xia, Xinfeng Xie, Xuchao Jia, Xuewei Wang, Yaelle Goldschlag, Yashesh Gaur, Yasmine Babaei, Yi Wen, Yiwen Song, Yuchen Zhang, Yue Li, Yuning Mao, Zacharie Delpierre Coudert, Zheng Yan, Zhengxing Chen, Zoe Papakipos, Aaditya Singh, Aayushi Srivastava, Abha Jain, Adam Kelsey, Adam Shajnfeld, Adithya Gangidi, Adolfo Victoria, Ahuva Goldstand, Ajay Menon, Ajay Sharma, Alex Boesenberg, Alexei Baevski, Allie Feinstein,

Amanda Kallet, Amit Sangani, Amos Teo, Anam Yunus, Andrei Lupu, Andres Alvarado, Andrew Caples, Andrew Gu, Andrew Ho, Andrew Poulton, Andrew Ryan, Ankit Ramchandani, Annie Dong, Annie Franco, Anuj Goyal, Aparajita Saraf, Arkabandhu Chowdhury, Ashley Gabriel, Ashwin Bharambe, Assaf Eisenman, Azadeh Yazdan, Beau James, Ben Maurer, Benjamin Leonhardi, Bernie Huang, Beth Loyd, Beto De Paola, Bhargavi Paranjape, Bing Liu, Bo Wu, Boyu Ni, Braden Hancock, Bram Wasti, Brandon Spence, Brani Stojkovic, Brian Gamido, Britt Montalvo, Carl Parker, Carly Burton, Catalina Mejia, Ce Liu, Changan Wang, Changkyu Kim, Chao Zhou, Chester Hu, Ching-Hsiang Chu, Chris Cai, Chris Tindal, Christoph Feichtenhofer, Cynthia Gao, Damon Civin, Dana Beaty, Daniel Kreymer, Daniel Li, David Adkins, David Xu, Davide Testuggine, Delia David, Devi Parikh, Diana Liskovich, Didem Foss, Dingkan Wang, Duc Le, Dustin Holland, Edward Dowling, Eissa Jamil, Elaine Montgomery, Eleonora Presani, Emily Hahn, Emily Wood, Eric-Tuan Le, Erik Brinkman, Esteban Arcaute, Evan Dunbar, Evan Smothers, Fei Sun, Felix Kreuk, Feng Tian, Filippas Kokkinos, Firat Ozgenel, Francesco Caggioni, Frank Kanayet, Frank Seide, Gabriela Medina Florez, Gabriella Schwarz, Gada Badeer, Georgia Swee, Gil Halpern, Grant Herman, Grigory Sizov, Guangyi, Zhang, Guna Lakshminarayanan, Hakan Inan, Hamid Shojanazeri, Han Zou, Hannah Wang, Hanwen Zha, Haroun Habeeb, Harrison Rudolph, Helen Suk, Henry Aspegren, Hunter Goldman, Hongyuan Zhan, Ibrahim Damla, Igor Molybog, Igor Tufanov, Ilias Leontiadis, Irina-Elena Veliche, Itai Gat, Jake Weissman, James Geboski, James Kohli, Janice Lam, Japhet Asher, Jean-Baptiste Gaya, Jeff Marcus, Jeff Tang, Jennifer Chan, Jenny Zhen, Jeremy Reizenstein, Jeremy Teboul, Jessica Zhong, Jian Jin, Jingyi Yang, Joe Cummings, Jon Carvill, Jon Shepard, Jonathan McPhie, Jonathan Torres, Josh Ginsburg, Junjie Wang, Kai Wu, Kam Hou U, Karan Saxena, Kartikay Khandelwal, Katayoun Zand, Kathy Matosich, Kaushik Veeraraghavan, Kelly Michelena, Keqian Li, Kiran Jagadeesh, Kun Huang, Kunal Chawla, Kyle Huang, Lailin Chen, Lakshya Garg, Lavender A, Leandro Silva, Lee Bell, Lei Zhang, Liangpeng Guo, Licheng Yu, Liron Moshkovich, Luca Wehrstedt, Madian Khabza, Manav Avalani, Manish Bhatt, Martynas Mankus, Matan Hasson, Matthew Lennie, Matthias Reso, Maxim Groshev, Maxim Naumov, Maya Lathi, Meghan Keneally, Miao Liu, Michael L. Seltzer, Michal Valko, Michelle Restrepo, Mihir Patel, Mik Vyatskov, Mikayel Samvelyan, Mike Clark, Mike Macey, Mike Wang, Miquel Jubert Hermoso, Mo Metanat, Mohammad Rastegari, Munish Bansal, Nandhini Santhanam, Natascha Parks, Natasha White, Navyata Bawa, Nayan Singhal, Nick Egebo, Nicolas Usunier, Nikhil Mehta, Nikolay Pavlovich Laptev, Ning Dong, Norman Cheng, Oleg Chernoguz, Olivia Hart, Omkar Salpekar, Ozlem Kalinli, Parkin Kent, Parth Parekh, Paul Saab, Pavan Balaji, Pedro Rittner, Philip Bontrager, Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanchandani, Pritish Yuvraj, Qian Liang, Rachad Alao, Rachel Rodriguez, Rafi Ayub, Raghotham Murthy, Raghu Nayani, Rahul Mitra, Rangaprabhu Parthasarathy, Raymond Li, Rebekkah Hogan, Robin Battey, Rocky Wang, Russ Howes, Rutu Rinott, Sachin Mehta, Sachin Siby, Sai Jayesh Bondu, Samyak Datta, Sara Chugh, Sara Hunt, Sargun Dhillon, Sasha Sidorov, Satadru Pan, Saurabh Mahajan, Saurabh Verma, Seiji Yamamoto, Sharadh Ramaswamy, Shaun Lindsay, Shaun Lindsay, Sheng Feng, Shenghao Lin, Shengxin Cindy Zha, Shishir Patil, Shiva Shankar, Shuqiang Zhang, Shuqiang Zhang, Sinong Wang, Sneha Agarwal, Soji Sajuyigbe, Soumith Chintala, Stephanie Max, Stephen Chen, Steve Kehoe, Steve Satterfield, Sudarshan Govindaprasad, Sumit Gupta, Summer Deng, Sungmin Cho, Sunny Virk, Suraj Subramanian, Sy Choudhury, Sydney Goldman, Tal Remez, Tamar Glaser, Tamara Best, Thilo Koehler, Thomas Robinson, Tianhe Li, Tianjun Zhang, Tim Matthews, Timothy Chou, Tzook Shaked, Varun Vontimitta, Victoria Ajayi, Victoria Montanez, Vijai Mohan, Vinay Satish Kumar, Vishal Mangla, Vlad Ionescu, Vlad Poenaru, Vlad Tiberiu Mihailescu, Vladimir Ivanov, Wei Li, Wenchen Wang, Wenwen Jiang, Wes Bouaziz, Will Constable, Xiaocheng Tang, Xiaoqian Wu, Xiaolan Wang, Xilun Wu, Xinbo Gao, Yaniv Kleinman, Yanjun Chen, Ye Hu, Ye Jia, Ye Qi, Yenda Li, Yilin Zhang, Ying Zhang, Yossi Adi, Youngjin Nam, Yu, Wang, Yu Zhao, Yuchen Hao, Yundi Qian, Yunlu Li, Yuzi He, Zach Rait, Zachary DeVito, Zef Rosnbrick, Zhaoduo Wen, Zhenyu Yang, Zhiwei Zhao, and Zhiyu Ma. The llama 3 herd of models, 2024.

- [8] Xu Han, Yulin Zhang, Ning Ding, et al. A survey on parameter-efficient fine-tuning of large language models. arXiv preprint arXiv:2402.05145, 2024.
- [9] Junxian He et al. On the effectiveness of parameter-efficient fine-tuning. arXiv preprint arXiv:2112.xxxxx, 2021.
- [10] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-efficient transfer learning for nlp. arXiv preprint arXiv:1902.00751, 2019.
- [11] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021.

- [12] Donghwan Jang et al. Why lora resists label noise: A theoretical framework for noise-robust parameter-efficient fine-tuning. arXiv preprint, 2024.
- [13] Damjan Kalajdzievski. A rank stabilization scaling factor for fine-tuning with lora, 2023.
- [14] Mahdi Karimi et al. Compelling properties of parameter-efficient fine-tuning. arXiv preprint arXiv:2110.xxxxx, 2021.
- [15] Anastasis Kratsios, Tin Sum Cheng, Aurelien Lucchi, and Hartz Sáez de Ocáriz Borde. Sharp generalization bounds for foundation models with asymmetric randomized low-rank adapters, 2025.
- [16] Brian Lester, Rami Al-Rfou, and Noah Constant. The power of scale for parameter-efficient prompt tuning. In Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, pages 3045–3059, 2021.
- [17] Xiang Lisa Li and Percy Liang. Prefix-tuning: Optimizing continuous prompts for generation. arXiv preprint arXiv:2101.00190, 2021.
- [18] Kai Lion, Liang Zhang, Bingcong Li, and Niao He. Polar: Polar-decomposed low-rank adapter representation. arXiv preprint arXiv:2506.03133, 2025.
- [19] Hao Liu et al. Convergence analysis of lora fine-tuning. arXiv preprint arXiv:2311.xxxxx, 2023.
- [20] Shih-Yang Liu, Chien-Yi Wang, Hongxu Yin, Pavlo Molchanov, Yu-Chiang Frank Wang, Kwang-Ting Cheng, and Min-Hung Chen. Dora: Weight-decomposed low-rank adaptation, 2024.
- [21] Shih-Yang Liu, Chien-Yi Wang, Hongxu Yin, Pavlo Molchanov, Yu-Chiang Frank Wang, Kwang-Ting Cheng, and Min-Hung Chen. Lora+: Efficient low rank adaptation of large models. arXiv preprint arXiv:2402.12354, 2024.
- [22] Grigory Malinovsky et al. Randomized asymmetric chain of lora: The first meaningful theoretical framework for low-rank adaptation. arXiv preprint arXiv:2410.08305, 2024.
- [23] David McAllester. Simplified pac-bayesian margin bounds. In Learning Theory and Kernel Machines: 16th Annual Conference on Learning Theory and 7th Kernel Workshop, COLT/Kernel 2003, Washington, DC, USA, August 24-27, 2003. Proceedings, pages 203–215. Springer, 2003.
- [24] David A McAllester. Pac-bayesian model averaging. In Proceedings of the twelfth annual conference on Computational learning theory, pages 164–170, 1999.
- [25] Fanxu Meng, Zhaohui Wang, and Muhan Zhang. Pissa: Principal singular values and singular vectors adaptation of large language models, 2025.
- [26] Takeru Miyato, Toshiki Kataoka, Masanori Koyama, and Yuichi Yoshida. Spectral normalization for generative adversarial networks. arXiv preprint arXiv:1802.05957, 2018.
- [27] Behnam Neyshabur, Srinadh Bhojanapalli, David McAllester, and Nati Srebro. A pac-bayesian approach to spectrally-normalized margin bounds for neural networks. arXiv preprint arXiv:1706.08947, 2017.
- [28] Behnam Neyshabur, Ryota Tomioka, and Nathan Srebro. Norm-based capacity control in neural networks. arXiv preprint arXiv:1503.00036, 2015.
- [29] Yujia Qin, Zhiyuan Hu, Yankai Lin, Wanxiang Che, Ning Ding, and Xuanjing Huang. Towards a unified view of parameter-efficient transfer learning. arXiv preprint arXiv:2110.04366, 2022.
- [30] Sylvestre-Alvise Rebuffi, Alexander Kolesnikov, Georg Sperl, and Christoph H Lampert. Learning multiple visual domains with residual adapters. In Advances in Neural Information Processing Systems, volume 30, 2017.
- [31] Mark Rudelson and Roman Vershynin. Sampling from large matrices: An approach through geometric functional analysis. Journal of the ACM (JACM), 54(4):21–es, 2007.
- [32] Thiziri Nait Saada, Alireza Naderi, and Jared Tanner. Mind the gap: a spectral analysis of rank collapse and signal propagation in attention layers, 2025.
- [33] John Shawe-Taylor and Robert C Williamson. Pac-bayesian bounds for classification and regression. In Proceedings of the 10th Annual Conference on Computational Learning Theory, pages 1–10, 1997.
- [34] Kimi Team, Yifan Bai, Yiping Bao, Guanduo Chen, Jiahao Chen, Ningxin Chen, Ruijue Chen, Yanru Chen, Yuankun Chen, Yutian Chen, Zhuofu Chen, Jiale Cui, Hao Ding, Mengnan Dong, Angang Du, Chenzhuang Du, Dikang Du, Yulun Du, Yu Fan, Yichen Feng, Keli Fu, Bofei Gao, Hongcheng Gao, Peizhong Gao, Tong Gao, Xinran Gu, Longyu Guan, Haiqing Guo, Jianhang Guo, Hao Hu, Xiaoru Hao, Tianhong He, Weiran He, Wenyang He, Chao Hong, Yangyang Hu, Zhenxing Hu, Weixiao Huang, Zhiqi

- Huang, Zihao Huang, Tao Jiang, Zhejun Jiang, Xinyi Jin, Yongsheng Kang, Guokun Lai, Cheng Li, Fang Li, Haoyang Li, Ming Li, Wentao Li, Yanhao Li, Yiwei Li, Zhaowei Li, Zheming Li, Hongzhan Lin, Xiaohan Lin, Zongyu Lin, Chengyin Liu, Chenyu Liu, Hongzhang Liu, Jingyuan Liu, Junqi Liu, Liang Liu, Shaowei Liu, T. Y. Liu, Tianwei Liu, Weizhou Liu, Yangyang Liu, Yibo Liu, Yiping Liu, Yue Liu, Zhengying Liu, Enzhe Lu, Lijun Lu, Shengling Ma, Xinyu Ma, Yingwei Ma, Shaoguang Mao, Jie Mei, Xin Men, Yibo Miao, Siyuan Pan, Yebo Peng, Ruoyu Qin, Bowen Qu, Zeyu Shang, Lidong Shi, Shengyuan Shi, Feifan Song, Jianlin Su, Zhengyuan Su, Xinjie Sun, Flood Sung, Heyi Tang, Jiawen Tao, Qifeng Teng, Chensi Wang, Dinglu Wang, Feng Wang, Haiming Wang, Jianzhou Wang, Jiaping Wang, Jinhong Wang, Shengjie Wang, Shuyi Wang, Yao Wang, Yejie Wang, Yiqin Wang, Yuxin Wang, Yuzhi Wang, Zhaoji Wang, Zhengtao Wang, Zhexu Wang, Chu Wei, Qianqian Wei, Wenhao Wu, Xingzhe Wu, Yuxin Wu, Chenjun Xiao, Xiaotong Xie, Weimin Xiong, Boyu Xu, Jing Xu, Jinjing Xu, L. H. Xu, Lin Xu, Suting Xu, Weixin Xu, Xinran Xu, Yangchuan Xu, Ziyao Xu, Junjie Yan, Yuzi Yan, Xiaofei Yang, Ying Yang, Zhen Yang, Zhilin Yang, Zonghan Yang, Haotian Yao, Xingcheng Yao, Wenjie Ye, Zhuorui Ye, Bohong Yin, Longhui Yu, Enming Yuan, Hongbang Yuan, Mengjie Yuan, Haobing Zhan, Dehao Zhang, Hao Zhang, Wanlu Zhang, Xiaobin Zhang, Yangkun Zhang, Yizhi Zhang, Yongting Zhang, Yu Zhang, Yutao Zhang, Yutong Zhang, Zheng Zhang, Haotian Zhao, Yikai Zhao, Huabin Zheng, Shaojie Zheng, Jianren Zhou, Xinyu Zhou, Zaida Zhou, Zhen Zhu, Weiye Zhuang, and Xinxing Zu. Kimi k2: Open agentic intelligence, 2025.
- [35] Yuandong Tian et al. Optimization dynamics of low-rank adaptation. arXiv preprint arXiv:2405.xxxxx, 2024.
- [36] Joel A Tropp. User-friendly tail bounds for sums of random matrices. *Foundations of Computational Mathematics*, 12(4):389–434, 2012.
- [37] Joel A Tropp et al. An introduction to matrix concentration inequalities. *Foundations and Trends in Machine Learning*, 8(1-2):1–230, 2015.
- [38] Mojtaba Valipour, Maysam You, Mehdi Karami, Ivan Osipov, Tushar Kalluri, Sanmi Koyejo, et al. Dylora: Parameter-efficient tuning of pre-trained models using dynamic search-free low-rank adaptation. arXiv preprint arXiv:2210.07558, 2023.
- [39] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need, 2023.
- [40] Roman Vershynin. High-dimensional probability: An introduction with applications in data science, volume 47. Cambridge university press, 2018.
- [41] Alexandre Virmaux and Kevin Scaman. Lipschitz regularity of deep neural networks: analysis and efficient estimation of the lipschitz constant. *Advances in Neural Information Processing Systems*, 31, 2018.
- [42] Yuxin Wang et al. Understanding the mechanism of lora optimization. arXiv preprint arXiv:2404.xxxxx, 2024.
- [43] Xiaoyu Wu et al. On the generalization of low-rank adaptation. arXiv preprint arXiv:2403.xxxxx, 2024.
- [44] Lingjiao Xu, Yujia Zhang, et al. Parameter-efficient fine-tuning methods for pretrained language models: A critical review and assessment. arXiv preprint arXiv:2312.12148, 2023.
- [45] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiayi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report, 2025.
- [46] Yuichi Yoshida and Takeru Miyato. Spectral norm regularization for improving the generalizability of deep learning. arXiv preprint arXiv:1705.10941, 2017.
- [47] Seungyong Yun et al. Transformers are rnns: Fast autoregressive transformers with linear attention. In *International Conference on Machine Learning*, 2019.
- [48] Yuchen Zeng and Kangwook Lee. The expressive power of low-rank adaptation, 2024.
- [49] Qingru Zhang, Minshuo Chen, Alexander Bukharin, Nikos Karampatziakis, Pengcheng He, Yu Cheng, Weizhu Chen, and Tuo Zhao. Adalora: Adaptive budget allocation for parameter-efficient fine-tuning. arXiv preprint arXiv:2303.10512, 2023.

- [50] Jiacheng Zhu, Kristjan Greenewald, Kimia Nadjahi, Haitz Sáez De Ocáriz Borde, Rickard Brüel Gabrielsson, Leshem Choshen, Marzyeh Ghassemi, Mikhail Yurochkin, and Justin Solomon. Asymmetry in low-rank adapters of foundation models. arXiv preprint arXiv:2402.16842, 2024.